



Trace element geochemistry of sphalerite and chalcopyrite in arc-hosted VMS deposits

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ABSTRACT

Trace element compositions of sphalerite and chalcopyrite have been investigated for four arc-hosted Cretaceous VMS deposits (María Teresa, Perubar, and Palma in Peru, and Cerro de Maimón in the Dominican Republic) using laser ablation inductively-coupled plasma mass spectrometry. In sphalerite, Fe, Mn, Cd, Hg, Ag, Sb, Se, In, Ge, and Ga are lattice-bound, whereas Sn, Tl, Bi, and Pb occur at least partly as mineral microinclusions. Significant variations in the contents of minor and trace elements are observed in sphalerite grains from individual deposits. A strong negative correlation between Fe (Inter-Quartile Range [IQR] 44,009–18,168 ppm) and Zn indicates a dominant $\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$ simple isovalent substitution. Regarding potential *high-tech* by-products, the contents of In (IQR 42–1.6 ppm, up to 415 ppm) and Ga (IQR 31–5.9 ppm, up to 96 ppm) in the studied sphalerite are normally much higher than those of Ge (IQR 0.85–0.16 ppm, up to 9.3 ppm). Correlation trends suggest $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, and more complex substitution mechanisms of Zn involving combinations of monovalent (Cu^+ , Ag^+), divalent (Ge^{2+}), trivalent (In^{3+} , Ga^{3+} , $[\text{Sn}^{3+}]$), and quadrivalent (Ge^{4+} , Sn^{4+}) cations.

As for chalcopyrite, Zn, Ag, Sn, Cd, Se, In, Ga, and Ge are interpreted to be lattice-bound, whereas Mo, Au, Tl, Sb, Pb, and Bi probably occur as microinclusions. Relative to sphalerite, chalcopyrite is depleted in In (IQR 28–8.4 ppm, up to 49 ppm) and Ga (IQR 8.3–2.9 ppm, up to 24 ppm) and enriched in Ge (IQR 5.9–0.70 ppm, up to 80 ppm). Of the other trace elements, Zn (IQR 426–190 ppm) is the most highly concentrated in the studied chalcopyrite, followed by Ag (IQR 136–23 ppm), Se (IQR 64–22 ppm), Sn (IQR 53–1.3 ppm), and Cd (IQR 12–6.0 ppm). General positive correlation trends between Zn, Cd, In, Ge, and Ga in chalcopyrite suggest varied coupled substitution mechanisms of Fe and Cu with fluctuating valences due to covalent bonding.

Trace-element distribution patterns in sphalerite and chalcopyrite were studied for the zone-refined Sofia-D massive sulfide body in the María Teresa deposit, which comprises a lower lower portion of dominant pyrite sheathed upward by zones of chalcopyrite (Cu zone), sphalerite (Zn zone) and galena + sphalerite $\pm$ fahlore $\pm$ barite (Pb – Zn – Ag zone). Bottom to top of the sulfide body, sphalerite records progressive depletion in In, Cu, Mn, and Se, and enrichment in Ge. This distribution pattern agrees with increasing crystallization temperatures and/or volatile magmatic influx towards the lower portion of the massive mineralization. Distribution of trace elements in chalcopyrite is rather uneven except for a sustained enrichment in Se towards the basal portion of the sulfide body. The fact that such trends are preserved in spite of extensive recrystallization during thermal metamorphism in parts of the Sofia-D massive sulfide mineralization suggests i) a closed metamorphic system and ii) that element interdiffusion was prominently local. Accordingly, we propose that sphalerite lattice-bound trace elements distribution patterns described in this article can help determine the polarity of massive sulfide bodies in VMS districts in metamorphosed and tectonized terranes.

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1. Introduction

Volcanogenic massive sulfide (VMS) deposits are exhalative and/or replacive accumulations of sulfide minerals formed at or near the seafloor where uprising hydrothermal fluids driven and in part generated by synchronous magmatism are entrained with cold seawater or porewaters (Doyle and Allen, 2003; Shanks and Thurston, 2012; Yeats et al., 2014; Tornos et al., 2015). VMS mineralization occurs as sulfide mounds to tabular stratiform bodies, which are often underlain by a stockwork or feeder zone (Galley et al., 2007; Tornos et al., 2015). Their formation responds to the convergence in the time and space of anomalously high thermal gradients and extensional tectonic regimes that promote the circulation of metal-bearing magmatic fluids and/or leaching of metals from the oceanic strata by evolved hydrothermal fluids (Ohmoto, 1996; Tivey, 2007; de Ronde et al., 2011; Piercey, 2011). Such geological features are primarily met in oceanic spreading ridges, arc environments, and rifted continental margins.

Most VMS deposits are preserved in suprasubduction zones either in back-arc, intra-arc, or forearc basins on both oceanic and continental crust (Franklin et al., 1998; Nimis et al., 2003; Herrington et al., 2005; de Ronde et al., 2011; Hollis et al., 2014; Peter et al., 2014; Torró et al., 2016a). In arc-related systems, contrasting water depths in massive sulfide formation exert a major control on the ore and hydrothermal alteration mineral assemblages and the metalliferous contents (Sillitoe et al., 1996; Dubé et al., 2007; Monecke et al., 2014). VMS deposits in back-arc and forearc basins form at water depths typically in the range between 1500 and 3700 m, whereas those in arc volcanoes and arc-related rifts are normally restricted to water depths in the range between 700 and 2000 m. The latter are often associated with elevated magmatic gas contents and boiling of the hydrothermal fluids may occur in shallower arc volcanoes and rifted intra-arc submarine hydrothermal systems (Gemmell et al., 2004; Dubé et al., 2007; Monecke et al., 2014).

VMS deposits are important sources of Cu, Zn, Pb, Au, and Ag (Galley et al., 2007). In addition, a wealth of other metals and metalloids that are potentially economically recoverable as byproducts (e.g., Sn, Co, Ni, In, Ga, Ge, Mo, Re, Se, Te) or deleterious (i.e., harmful or unwanted elements; e.g., Mn, As, Sb, Hg, Tl) may occur in remarkable concentrations in VMS deposits and even determine their economic viability (Benzaazoua et al., 2003; Zaccarini and Garuti, 2008; Koski and Mosier, 2012; Ridley, 2012; Carvalho et al., 2018; Frenzel et al., 2019). Minor and trace elements can either precipitate as native elements or substitute into the crystal lattice of major sulfide minerals (Cook et al., 2009, 2012, 2015; Ye et al., 2011; Ridley, 2012; Murakami and Ishihara, 2013; George et al., 2015, 2018; Benites et al., 2021a, 2021b). The understanding of the mineralogical expression of either potential by-product or deleterious minor and trace elements in a given deposit or mineralization style is essential to predict their metallurgical behavior (e.g., Cook et al., 2011).

This study aims at understanding the mineralogical expression of minor and trace elements in sphalerite and chalcopyrite from arc-related VMS deposits and their distribution across massive sulfide bodies. To do so, we have systematically determined the trace element budget on these ore minerals from four arc-related VMS deposits in the Andes (María Teresa, Perubar, and Palma in Peru) and the Caribbean (Cerro de Maimón in the Dominican Republic). The obtained contents are compared with published data on other VMS deposits worldwide. In addition, a comprehensive set of samples has allowed the study of ore body-scale zoning of certain trace elements in a zone-refined massive sulfide body in the María Teresa deposit. The obtained results are used to assess the incorporation mechanisms of some *high-tech* elements to the crystal lattice of major sulfides (sphalerite and chalcopyrite) and contribute to revealing element distribution patterns across massive sulfide bodies in VMS districts; such patterns can be used to target the most prospective zones of zoned massive sulfide bodies for potential by-products and complementarily also to constrain sulfide body polarity during exploration in tectonized terranes.

2. Geological setting

2.1. Volcanogenic massive sulfide deposits in Peru

2.1.1. General geologic framework

In Peru, several VMS deposits occur along the coastal region and are hosted by rocks of Cretaceous age (Steinmüller et al., 2000; Fontboté, 2019). In northern Peru, the giant Tambogrande deposits represent the largest VMS endowment in this country and probably in South America, and include the TG1 deposit with resources of 109 million metric tons (Mt) grading 1.6% Cu, 1.0% Zn, 0.5 g/t Au, and 22 g/t Ag, plus 16.7 Mt grading 3.5 g/t Au and 64 g/t Ag in oxide ore, and the TG3 deposit with resources of 82 Mt grading 1.0% Cu, 1.4% Zn, 0.8 g/t Au, and 25 g/t Ag (Manhattan Minerals, 2002 in Winter et al., 2010). There, VMS mineralization is hosted by subalkaline (calc-alkaline and tholeiitic) basaltic pillowed and brecciated lavas and lesser amounts of felsic volcanic rocks that filled N and NW-oriented sub-basins in the lower portion of the Lancones basin (Fig. 1; Tegart et al., 2000; Winter et al., 2004, 2010). The formation of the Lancones basin is connected to the break-up of Gondwana and coincided with a period of intense extension and rifting due to trench-rollback (Winter et al., 2010 and their references). Minimum age of massive sulfide formation (and likely synchronous with or slightly postdating massive sulfide formation) is constrained by U—Pb zircon dating of felsic volcanoclastic rocks at 104.8 ± 1.3 to 100.2 ± 0.5 Ma (middle Albian; Winter et al., 2010). The volcanic and sedimentary pile in the Lancones basin was intruded by Late Cretaceous to Tertiary arc-related granitoids of the Coastal Batholith (cf. Pitcher, 1985; Atherton, 1990; Cobbing, 1999; Hildebrand and Whalen, 2014).

In the central coastal segment of Peru, where the deposits studied in this paper are located, VMS mineralizations emplaced in the Huarney and Cañete basins of the West Peruvian Trough (WPT; Fig. 1) and are now exposed along the Peruvian Western Cordillera to the Coastal region. The opening of the WPT was assigned to an aborted back-arc or true ocean floor extensional marginal basin (e.g., Atherton et al., 1985; Aguirre et al., 1989; Atherton and Aguirre, 1992) and more recently has been interpreted as an intra-arc extensional pull-apart basin (Soler, 1991; Aillard, 1994; Polliand et al., 2005; Polliand, 2006). Stratigraphic relationships in the WPT led Scherrenberg et al. (2012) to delineate i) a western intra-arc basin, which is dominated by igneous rocks and that geographically coincides with the VMS mineralization; and ii) an eastern, deeper back-arc basin in which volcanic rocks overlie continental clastic rocks and marine carbonates. Major basin filling in the WPT occurred from latest Jurassic to Late Cretaceous (Mégard, 1984; Benavides-Cáceres, 1999). The stratigraphy of the WPT (at least in its western portion *sensu* Scherrenberg et al., 2012) is largely composed of a volcano-sedimentary sequence including bimodal mafic-felsic volcanic rocks, limestones, shales, sandstones, and local evaporites that was intruded by the Peruvian Coastal Batholith, which in part is coetaneous with the volcano-sedimentary sequence (Mégard, 1984; Atherton and Aguirre, 1992; Vela, 1997; Benavides-Cáceres, 1999; Polliand et al., 2005; Polliand, 2006; Hildebrand and Whalen, 2014). The stratigraphy of the WPT is complex and includes abrupt lateral facies changes greatly limiting the stratigraphic correlation and age estimates of the sedimentary units. Since the scope of this article is limited to the geochemical composition of some ore minerals in selected VMS deposits, we will not elaborate on the nomenclature of hosting units but will simply conform with 'Casma Group' embracing the Albian-Cenomanian volcano-sedimentary sequence versus the younger 'Upper Cretaceous-Paleocene volcano-sedimentary sequence' (Romero et al., 2008; Cueva et al., 2010; Fontboté, 2019).

From north to south, the most representative VMS mineralization in the Huarney and Cañete basins are the María Teresa, Aurora Augusta, Perubar, Palma, and Cerro Lindo (Fig. 1). Geochronology has given two groups of ages for these deposits: uppermost Cretaceous and mid-Cretaceous. High-resolution U—Pb zircon geochronology performed by Polliand et al. (2005) on rhyolitic lavas at the bottom and top of

massive ore bodies in Perubar yields dates of 69.71 ± 0.18 and 68.92 ± 0.16 Ma, and on intrusive rocks, dates of 67.91 ± 0.17 and 67.89 ± 0.18 Ma. These dates ascribe the hosting rocks to the Upper Cretaceous-Paleocene volcano-sedimentary sequence and point to largely coeval subsidence, submarine volcanism, and plutonic activity within the Andean magmatic arc during the Late Cretaceous (Polliand et al., 2005; Polliand, 2006). At María Teresa, a date of 68 ± 6 Ma (Rb/Sr in hydrothermal sericite; Romero et al., 2008) points also to VMS

mineralization in the latest Cretaceous. In contrast, the Cerro Lindo VMS mineralization is described to be hosted by a volcano-sedimentary sequence whose age has been attributed to the mid-Cretaceous (Casma Group) according to U–Pb zircon dating of host rhyolites at 105.7 ± 0.2 and 106.9 ± 0.2 Ma (Garipey and Hinostroza, 2013; Votorantim, 2017; Bueno Carreón, 2019; Bueno Carreón and Mendoza Mondragón, 2019). Recent U–Pb zircon dating at 101 Ma of a diorite dike crosscutting the mineralization suggests that the Palma VMS deposit is also mid

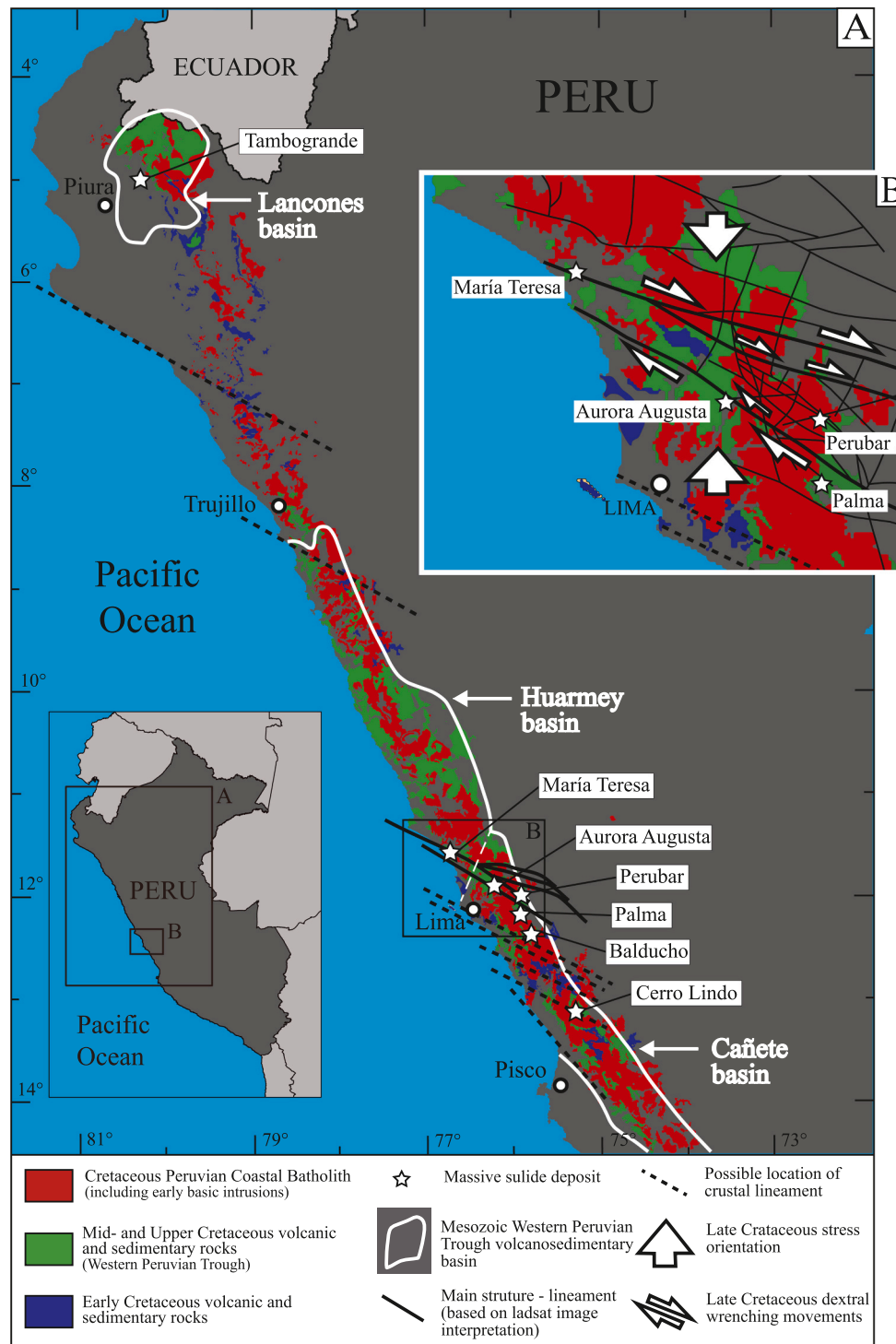


Fig. 1. Geologic map of the Peruvian Coastal Batholith and Mesozoic Western Peruvian Trough (WPT) with related volcano-sedimentary basins hosting massive sulfide deposits. Modified from Polliand et al. (2005). Distribution of the Cretaceous Peruvian Coastal Batholith and strata after the digital version of the geological map of Peru (INGEMMET, 2000).

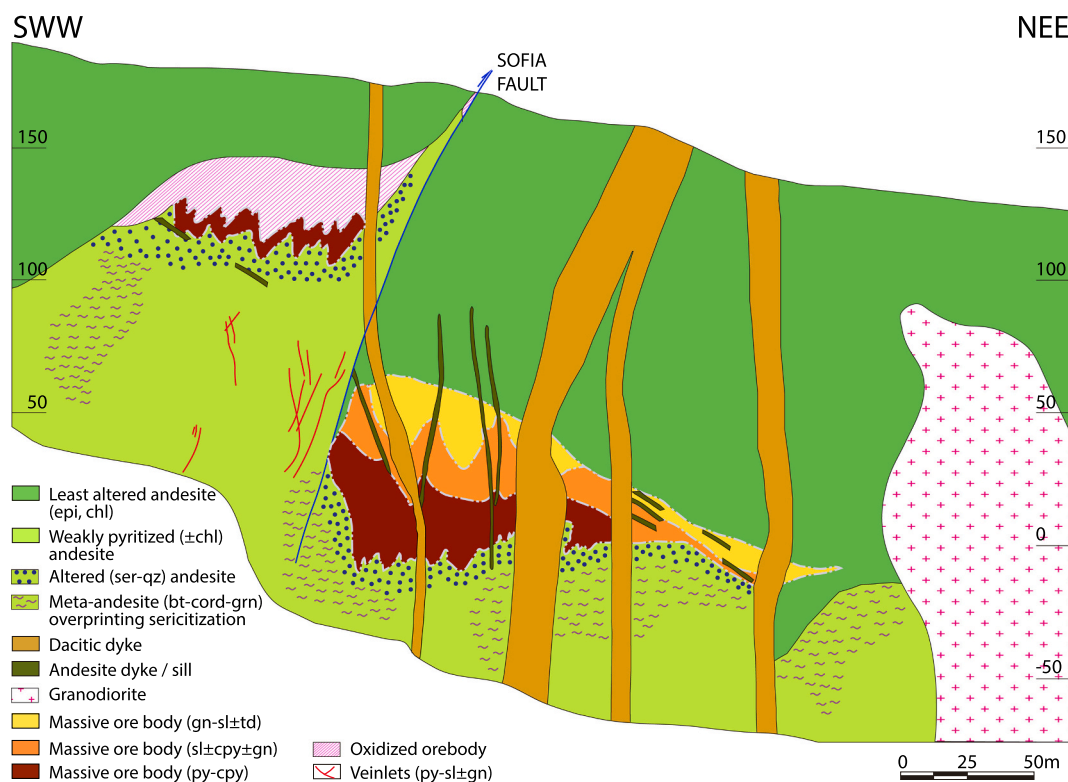


Fig. 2. Representative schematic cross-section across the Sofia D massive sulfide body in the María Teresa VMS deposit, Peru (redrawn from geological documents of the María Teresa Mine geological department as of December 2020).

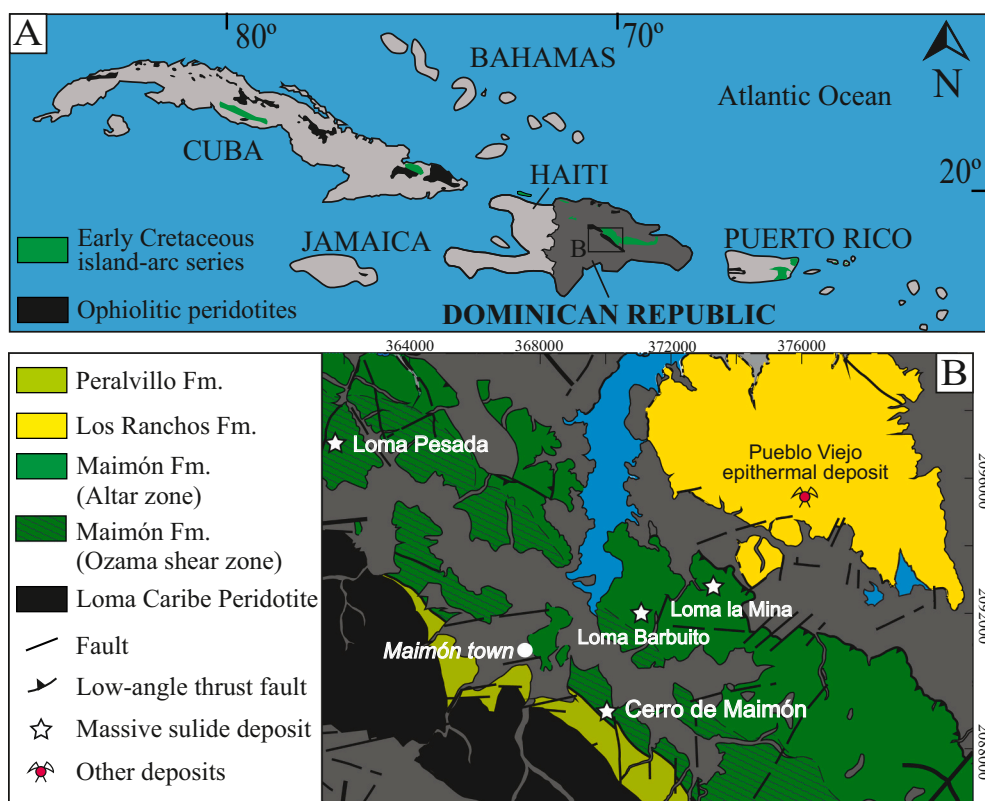


Fig. 3. (A) Geographic location of the Early Cretaceous boninite and island arc tholeiite volcanic series and ophiolitic peridotites along the Greater Antilles. (B) Geological map of the Maimón Formation and surrounding Early Cretaceous arc-related volcanic units (modified from Martín and Draper, 1999).

Table 1

Summary of element contents in sphalerite from the María Teresa, Perubar, Palma, and Cerro de Maimón VMS deposits (LA-ICP-MS data).

Deposit	Sector / Body / Section	Mn (ppm)	Fe (ppm)	Cu (ppm)	Zn (ppm)	Ga (ppm)	Ge (ppm)	As (ppm)	Se (ppm)	Mo (ppm)	Ag (ppm)	Cd (ppm)	In (ppm)	Sn (ppm)*	Sb (ppm)	Hg (ppm)	Tl (ppm)*	Pb (ppm)*	Bi (ppm)*
All samples (n = 141)	Min.	107	9772	3.8	569500	1.4	B.D.L.	4.6	B.D.L.	B.D.L.	2.2	4720	0.44	0.14	B.D.L.	98	B.D.L.	0.038	B.D.L.
	Max.	4998	63534	740	663300	94	2.3	9.3	32	1.6	66	9409	415	20	48	788	19	44	3.5
	GM	1353	29317	96	623168	25	0.51	6.0	6.4	0.16	11	6605**	60	1.2	4.3	392	1.0	6.2	0.44
	SD	1236	13930	106	22726	19	0.56	0.8	9.3	0.19	10	1279	89	2.8	7.2	135	2.1	7.2	0.77
	IQR	2343-326	40675-17122	98-36	641300-608100	32-11	0.65-0.17	6.5-5.5	5.3-1.6	0.20-0.047	12-4.2	7373-5290	68-4.4	0.61-0.25	5.2-0.52	448-326	1.2-0.060	8.7-1.4	0.60-0.0055
Bubulina / Bubulina 1180 (n = 9)	Min.	3139	13000	79	630300	17	B.D.L.	4.7	4.3	0.25	3.2	4720	21	0.30	0.048	664	B.D.L.	0.21	B.D.L.
	Max.	4998	18702	174	648500	94	0.16	5.8	5.3	0.73	5.2	5044	99	3.2	1.1	788	0.0081	4.3	B.D.L.
	GM	4474	16628	126	640091	51	0.16	5.3	4.8	0.37	4.2	4867	59	1.0	0.28	724	0.0067	1.2	B.D.L.
	SD	550	2135	26	6640	23	-	0.3	0.4	0.15	0.7	118	36	1.0	0.37	35	0.0010	1.7	B.D.L.
	IQR	4747-4492	18088-15614	138-118	643900-633600	68-43	0.16-0.16	5.4-5.1	5.0-4.6	0.36-0.31	4.6-3.8	4964-4804	94-25	1.7-0.35	0.54-0.15	737-704	0.0076-0.0061	4.1-0.62	B.D.L.
Calera / Sofia-D, all samples (n = 132)	Min.	107	9772	3.8	569500	1.4	B.D.L.	4.6	B.D.L.	B.D.L.	2.2	4896	0.44	0.14	B.D.L.	98	B.D.L.	0.038	B.D.L.
	Max.	2857	63534	740	663300	70	2.3	9.3	32	1.6	66	9409	415	20	48	677	19	44	3.46
	GM	716	26558	61	621589	17	0.33	6.0	3.1	0.09	7.9	6613	19	0.48	1.7	349	0.28	3.4	0.04
	SD	939	13984	109	22982	17	0.56	0.8	9.6	0.18	10.6	1235	91	2.86	7.3	107	2.18	7.3	0.77
	IQR	5771-303	40842-17173	87-36	639625-605100	31-11	0.66-0.17	6.5-5.5	5.5-1.5	0.17-0.045	13-4.2	7411-5446	54-4.3	0.58-0.24	5.5-0.65	438-318	1.3-0.084	8.8-1.6	0.60-0.0055
Calera / Sofia-D / 300 (n = 105)	Min.	107	9772	20	569500	3.1	B.D.L.	5.2	B.D.L.	B.D.L.	2.2	4896	3.9	0.14	B.D.L.	98	0.0072	0.038	B.D.L.
	Max.	2857	63534	740	663300	38	2.3	9.3	32	1.6	66	8051	415	16	48	485	19	33	3.5
	GM	828	25631	66	621999	15	0.37	6.1	3.3	0.11	7.5	6334	33	0.37	2.0	345	0.35	3.9	0.053
	SD	967	14897	120	24203	10	0.60	0.7	10.6	0.19	10.5	967	98	2.19	7.7	97	2.39	6.5	0.801
	IQR	2343-251	42100-15748	90-36	639600-602500	27-10	0.79-0.17	6.4-5.7	5.6-1.5	0.20-0.050	13-3.9	7239-5290	78-12	0.48-0.22	5.9-0.79	438-315	1.7-0.13	9.0-2.0	0.72-0.0057
Calera / Sofia D / 340 (n = 10)	Min.	113	18062	3.8	591600	1.4	B.D.L.	6.7	B.D.L.	B.D.L.	4.1	5592	0.44	0.18	0.10	213	B.D.L.	0.28	B.D.L.
	Max.	1662	42132	76	642900	38	0.21	7.1	7.1	0.13	9.2	7044	42	1.8	1.1	443	0.32	2.5	0.0051
	GM	468	31839	19	614743	8	0.16	6.9	3.1	0.06	6.9	6198	5	0.46	0.40	300	0.05	0.83	0.0046
	SD	799	9365	35	22717	19	0.04	0.1	2.8	0.05	1.8	656	22	0.45	0.39	105	0.12	0.74	0.0006
	IQR	1646-139	41573-27114	70-5.3	634275-594450	38-1.5	0.10-0.13	7.0-6.9	6.6-1.3	0.11-0.027	8.6-5.7	6899-5632	42-0.56	0.58-0.29	0.87-0.24	409-219	0.17-0.019	1.5-0.48	0.0050-0.0044
Calera / Sofia D / 500 (n = 5)	Min.	320	15296	40	608000	22	0.12	4.9	1.3	0.033	11	8905	1.1	0.34	0.31	329	0.085	3.9	B.D.L.
	Max.	584	38233	82	645800	59	0.95	5.6	3.7	0.61	46	9409	1.8	2.2	2.4	677	0.71	44	0.0061
	GM	502	19140	53	637688	28	0.37	5.3	2.1	0.07	24	9121	1.6	0.95	3.3	574	0.23	10	0.0061
	SD	97	8832	15	14717	14	0.31	0.2	0.9	0.23	14	166	0.2	0.71	9.6	130	0.24	15	-
	IQR	557-520	17392-16548	55-47	645150-641800	25-24	0.64-0.29	5.4-5.3	2.4-1.7	0.065-0.038	38-18	9152-9054	17-1.6	1.6-0.64	13-1.4	654-620	0.44-0.12	16-6.0	0.0061-0.0061
Calera / Sofia D / 540 (n = 12)	Min.	214	35835	82	608000	58	B.D.L.	4.6	2.1	0.037	5.7	8644	1.0	0.61	0.02	325	0.014	0.10	B.D.L.
	Max.	357	38973	137	619700	70	1.2	5.4	3.7	0.67	27	9168	1.5	20	11	356	1.2	31	0.011
	GM	322	37824	91	614685	64	0.22	5.0	2.8	0.06	8.9	8877	1.2	3.1	0.9	338	0.14	1.7	0.0069
	SD	36	1036	16	2950	4	0.33	0.3	0.4	0.18	6.2	158	0.2	5.6	4.0	11	0.45	8.6	0.0035
	IQR	339-326	38681-37310	89-84	616125-613700	68-63	0.23-0.16	5.2-4.9	3.0-2.6	0.055-0.042	11-6.5	8951-8784	1.2-1.1	8.4-1.1	3.1-0.28	344-331	0.38-0.052	4.7-0.45	0.0084-0.0053
All samples (n = 50)	min	667	30747	16	549400	3.4	B.D.L.	4.0	B.D.L.	0.046	1.7	3977	0.0056	0.17	B.D.L.	15	B.D.L.	0.038	B.D.L.
	max	7639	73876	789	617700	33	0.49	6.1	10	0.55	14	7076	7.1	1.4	5.4	117	1.0	17	0.0037
	GM	2421	151935	35	579909.2969	13	0.17	4.6	3.5	0.19	4.6	5252	1.0	0.27	0.45	44	0.05	1.1	0.0033
	SD	2738	15233	108	24068.28288	11	0.08	0.4	3.0	0.17	2.7	1070	1.8	0.17	1.15	39	0.18	3.3	0.0005
	IQR	5964-907	67570-39965	41-25	600000-551800	30-4.7	0.19-0.15	4.8-4.4	8.2-2.5	0.42-0.089	7.1-3.6	6664-9.42966	2.3-0.78	0.31-0.22	1.2-0.18	107-18	0.13-0.025	2.8-0.39	0.0035-0.0031
Cecilia Sur (n = 22)	Min.	667	30747	20	599800	11	B.D.L.	4.3	B.D.L.	0.046	4.3	5485	0.0056	0.20	B.D.L.	71	0.013	0.038	B.D.L.
	Max.	1047	42636	789	617700	33	0.19	5.4	4.4	0.20	14	7076	0.82	0.43	4.0	117	1.0	17	B.D.L.
	GM	906	37667	45	605057	25	0.16	4.7	1.7	0.085	6.8	6422	0.10	0.31	0.6	97	0.10	2.5	B.D.L.
	SD	75	3821	160	7381	7	0.03	0.3	1.3	0.031	2.2	518	0.39	0.07	1.1	16	0.23	4.1	B.D.L.
	IQR	939-879	40139-34273	46-36	614400-600000	31-20	0.17-0.14	4.8-4.6	3.2-1.1	0.098-0.071	7.8-5.7	6797-5833	0.75-0.0060	0.35-0.26	1.9-0.23	108-79	0.20-0.047	5.8-1.2	B.D.L.
Graciela (n = 10)	Min.	6704	59164	23	572900	17	B.D.L.	4.0	3.2	0.39	1.7	4799	1.9	0.37	0.59	43	0.17	3.2	B.D.L.
	Max.	7639	66451	30	575100	22	0.20	4.7	4.2	0.55	2.8	5135	2.1	0.23	0.59	43	0.17	3.2	B.D.L.
	GM	7267	63460	26	574430	21	0.18	4.3	3.7	0.48	2.0	4983	2.0	0.20	0.13	40	0.076	0.56	B.D.L.
	SD	240	2010	2	699	2	0.01	0.2	0.3	0.04	0.3	89	0.0	0.02	0.20	2	0.043	0.93	B.D.L.
	IQR	7367-7197	64413-62864	27-24	575100-574200	22-21	0.19-0.18	4.3-4.2	4.0-3.5	0.49-0.46	2.1-1.9	5039-4952	2.0-2.0	0.21-0.19	0.27-0.053	41-39	0.10-0.055	1.2-0.22	B.D.L.
Juanita (n = 4)	Min.	1669	58263	20	557900	4.5	B.D.L.	4.7	2.5	0.12	3.6	3977	6.7	0.19	0.38	15	0.0060	0.63	B.D.L.
	Max.	1973	64193	126	567000	8.1	0.49	6.1	3.9	0.17	7.4	4388	7.1	1.4	3.0	17	0.49	1.3	B.D.L.
	GM	1801	61300	41	562010	6.5	0.25	5.2	3.1	0.14	4.6	4388	6.9	0.35	0.91	16	0.05	0.83	B.D.L.
	SD	128	2706	49	4813	1.5	0.26	0.6	0.6	0.03	1.8	185	0.2	0.57	1.19	1	0.23	0.31	B.D.L.
	IQR	1856-1736	63248-59558	61-25	565725-557900	7.4-6.4	0.40-0.22	5.4-5.0	3.3-2.9	0.16-0.12	5.2-3.6	4341-4110	7.0-6.9	0.56-0.20	1.4-0.59	17-16	0.18-0.031	1.0-0.64	B.D.L.
Rimac-D (n = 14)	Min.	5121	66771	16	549400	3.4	B.D.L.	4.4	6.5	0.33	2.5	3996	2.1	0.19	0.14	17	B.D.L.	0.22	B.D.L.
	Max.	6090	73876	66	552400	4.8	0.22	4.9	10	0.45	10	4436	2.4	0.36	5.4	20	0.12	2.7	0.0037
	GM	5631	71109	27	551092	4.2	0.17	4.6	8.6	0.38	4.4	4235	2.3	0.25	0.47	18	0.011	0.52	0.0033
	SD	325	1984	13	1010	0.5	0.03	0.2	1.1	0.04	2.1	104	0.1	0.04	1.40	1	0.034	0.73	0.0005
	IQR	5915-5408	72557-70501	30-22	551600-550625	4.6-4.0	0.20-0.15	4.8-4.6	9.3-8.3	0.42-0.35	4.1-3.9	4291-4185	2.3-2.2	0.27-0.22	0.75-0.20	19-18	0.028-0.0		

Table 2

Summary of element contents in chalcopyrite from the María Teresa and Cerro de Maimón VMS deposits (LA-ICP-MS data).

Deposit	Sector/body/section		Fe (ppm)	Co (ppm)	Ni (ppm)	Cu (ppm)	Zn (ppm)	Ga (ppm)	Ge (ppm)	As (ppm)	Se (ppm)
María Teresa (Peru)	All samples (n = 40)	Min.	264,304	B.D.L.	B.D.L.	340,000	152	0.070	0.18	0.18	10
		Max.	373,972	0.087	2.2	352,700	647	24	80	1.9	59
		GM	331,188	0.075	1.5	346,217	277	4.0	1.6	0.43	25
		SD	24,272	0.010	0.4	3871	133	6.4	19.4	0.41	13
		IQR	348,538–319,792	0.079–0.070	1.9–1.2	348,300–343,900	425–194	9.2–2.6	4.3–0.66	0.74–0.29	31–19
	Calera/Sofía D/300 (n = 25)	Min.	264,304	B.D.L.	B.D.L.	342,000	152	0.070	0.18	0.18	10
		Max.	373,972	0.087	2.2	352,700	529	24	80	1.9	45
		GM	331,249	0.075	1.4	347,222	263	3.5	1.2	0.45	22
		SD	25,625	0.010	0.4	3155	124	6.8	20.9	0.43	8
		IQR	348,538–319,792	0.079–0.070	1.8–1.2	349,700–345,000	425–192	8.1–2.6	2.2–0.64	0.80–0.29	30–16
	Calera/Sofía D/500 (n = 4)	Min.	312,109	B.D.L.	B.D.L.	340,000	276	8.9	7.3	0.25	49
		Max.	350,244	B.D.L.	1.9	340,000	647	11	12	0.36	59
		GM	330,803	–	1.9	340,000	374	10	9.4	0.30	53
		SD	15,609	–	0.1	–	170	1	2.4	0.05	4
		IQR	336,816–325,247	–	1.9–1.8	340,000–340,000	425–303	10–9.1	11–7.8	0.33–0.28	54–50
	Cerro de Maimón (Dominican Republic) (n = 11)	Min.	270,450	1.2	B.D.L.	346,300	B.D.L.	3.1	4.8	0.36	60
		Max.	316,435	3.1	2.5	346,300	71	3.9	6.6	2.0	94
		GM	296,657	1.8	1.7	346,300	71	3.4	5.6	0.86	81
		SD	14,668	0.6	0.4	–	–	0.2	0.6	0.54	11
		IQR	307,234–285,613	2.0–1.5	1.9–1.5	346,300–346,300	–	3.6–3.3	6.1–5.3	1.3–0.56	89–79

B.D.L.: Below detection limit; IQR: interquartile range; GM: geometric mean; SD: standard deviation.

aRagged ablation spectra indicate that these elements may occur as sub-microinclusions.

volcanic and subvolcanic rocks that were intruded by a series of sub-vertical, N150E-oriented andesitic and dacitic dykes and, in places, by granodiorite and diorite plutons. Volcanic rocks at María Teresa comprise basaltic andesites and andesites and less abundant rhyodacites which, akin plutonic rocks in the district, show calc-alkaline affinity (Pichardo et al., 2019). Basaltic andesites and andesites show aphanitic, porphyritic textures drawn by plagioclase and amphibole phenocrysts. Amygdaloidal and brecciated textures are also common and, towards the upper section of the sequence, pillow lavas and hyaloclastites occur. Despite that the limit between individual lava flows is mostly blurred, Pichardo et al. (2019) conclude that the locally observed bedding is horizontal to slightly W-dipping. The N160E-oriented El Abra strike-slip fault dismembered and displaced clockwise the mineralization in the 'prospective horizon' so that the Calera sector has been displaced about 500 m to the south in relation to the Mine 2 and Bubulina sectors. In addition, the N140E-oriented Sofía reverse fault has slid the western block on top of the eastern block with an approximate vertical displacement of 90 m (Fig. 2). The mineralization in the María Teresa deposit occurs mostly as N150E-oriented irregular to lenticular massive sulfide bodies and underlying stockwork structures associated with abundant alteration to sericite and quartz (Pichardo et al., 2019). Close to the contact with granodiorite intrusions chiefly to the east of the Calera zone, a mineral assemblage including cordierite and garnet porphyroblasts along with biotite, phlogopite, and tremolite is interpreted as the contact metamorphic product of the hydrothermal sericite alteration (Pichardo et al., 2019). The ore mineral assemblage includes sphalerite, chalcopyrite, tetrahedrite-group minerals, and galena, which occur along with variable proportions of pyrite $\pm$ pyrrhotite $\pm$ barite. The Sofía-D massive sulfide body is the largest in the deposit (~8 Mt grading 7.1% Zn, 1.37% Cu, 0.64% Pb, 60.3 g/t Ag; Pichardo et al., 2019) and shows clear metal zoning (Fig. 2). Metal zoning, ore mineral assemblages, and textures observed in this ore body are described below. Due to the well-defined metal zoning in Sofía-D, it was selected in this study to trace the distribution of trace elements in sphalerite and chalcopyrite in zone-refined massive sulfide bodies.

2.1.3. Geology of the Perubar deposit

The Zn-Pb-Ba Perubar deposit is located 50 km E of Lima (Fig. 1) and was exploited between 1978 and 2000 with a production of 6 Mt with an average grade of 11.8% Zn and 1.4% Pb (Polliand, 2006). The massive sulfide mineralization is hosted by an uppermost Cretaceous volcano-sedimentary sequence composed of submarine volcanic rocks and

pyroclastic deposits of intermediate to felsic composition, intercalated with detrital sediments, volcanoclastic sandstones, tuffaceous mudstones, calcareous volcanoclastics, and limestones. Monzoniorite and monzogranite stocks belonging to the Coastal Batholith intruded in the area $< \sim 1$ Myr after the mineralizing event causing contact metamorphism at the Perubar deposit (Polliand et al., 1999; see Fig. 3 in Polliand et al., 2005; Polliand, 2006). The hydrothermal alteration of the host rocks comprises a siliceous core in the stockwork zone that grades outward to chlorite + pyrite $\pm$ sericite; extensive quartz + sericite developed in the footwall below the sulfide lenses, as well as distal quartz + K-feldspar. The deposit consists of four main massive barite-sulfide ore bodies (Graciela, Juanita, Cecilia Norte, and Cecilia Sur). The Theoretical (i.e., before tectonic deformation) ore zonation included a stockwork, which is now located north of the deposit, and massive mineralization that comprised pyrite(–magnetite) and pyrite-rich semi-black ores topped by a massive black ore, which graded laterally to banded barite-sulfide ores (Polliand, 2006). The ore included pyrrhotite, pyrite, marcasite, magnetite, sphalerite, galena, barite, and lesser proportions of chalcopyrite, tetrahedrite, and arsenopyrite (Polliand, 2006). In addition, during our survey, tiny amounts of molybdenite and electrum were identified (Fig. S1). Representative photomicrographs of the samples containing the analyzed sphalerite from Perubar are shown in Fig. S1. Polliand (2006) suggests that the conspicuously low proportion of chalcopyrite in the ore mineralization constrains the temperature of the mineralizing fluids $< \sim 300$ °C.

2.1.4. Geology of the Palma deposit

The Zn–Pb Palma deposit is located 55 km SE of Lima and 15 km south of Perubar (Fig. 1). Ongoing exploration by Volcan Cía. Minera since 2009 has revealed a resource of 14.5 Mt including 9.6 Mt indicated at 5% Zn, 0.7% Pb, and 22 g Ag/t and 4.9 Mt inferred at 6% Zn, 1% Pb, and 21 g Ag/t, all reported with a cut-off of 3.5% Zn (Farfán et al., 2019). The mineralization is hosted in a volcano-sedimentary sequence consisting of sills and NNW-SSE trending dykes of basaltic to andesitic composition and local peperitic textures, organic matter-rich and framboidal-pyrite-bearing pelites, and shallow-water limestones. This sequence is cut by NNW-SSE-oriented felsic dykes. The igneous suite in the district displays calc-alkaline affinity (Farfán et al., 2019). Hydrothermal alteration includes deep, poorly developed sericite alteration and much widespread albite + epidote + chlorite + pyrite alteration. The mineralization occurs as stacked massive sulfide bodies roughly parallel to the bedding and replaced chiefly limestone and subordinately pelitic

Mo (ppm) ^a	Ag (ppm)	Cd (ppm)	In (ppm)	Sn (ppm)	Sb (ppm) ^a	Au (ppm) ^a	Hg (ppm)	Tl (ppm) ^a	Pb (ppm) ^a	Bi (ppm) ^a
B.D.L.	3.2	3.5	11	4.3	1.4	B.D.L.	B.D.L.	0.019	1.4	0.0092
0.044	295	15	49	201	31	0.40	3.8	11	82	5.5
0.027	82	8.8	20	43	6.8	0.069	1.1	0.8	7.8	1.0
0.008	67	3.3	12	40	7.3	0.091	1.1	3.0	16.4	1.6
0.034–0.021	150–71	12–6.2	32–12	77–33	12–3.2	0.12–0.039	1.9–0.69	2.4–0.28	12–4.3	2.5–0.54
B.D.L.	3.2	3.5	11	4.3	1.4	B.D.L.	B.D.L.	0.019	1.4	0.0092
0.044	295	15	49	201	31	0.40	3.8	11	82	5.5
0.027	89	9.2	18	42	6.6	0.077	1.0	0.8	8.8	1.0
0.008	68	3.2	11	43	7.4	0.095	0.9	3.2	17.2	1.6
0.035–0.021	166–76	12–8.2	23–12	78–30	10–3.2	0.13–0.041	1.5–0.69	2.4–0.28	14–6.0	2.6–0.86
B.D.L.	44	4.1	34	43	3.6	0.016	0.57	0.62	1.5	0.45
0.033	68	9.6	44	53	17	0.075	3.7	3.4	7.9	2.5
0.025	52	6.6	37	49	7.8	0.036	1.9	1.7	3.6	1.0
0.007	10	2.4	5	4	7.2	0.029	1.5	1.1	3.2	1.0
0.028–0.022	55–48	8.4–5.4	38–34	52–47	16–3.8	0.065–0.020	3.5–1.5	2.5–1.5	7.0–1.9	1.8–0.52
5.4	8.3	B.D.L.	6.7	1.2	0.05	0.015	B.D.L.	0.026	5.1	1.5
250	33	0.59	8.7	1.8	0.34	0.13	0.27	0.42	39	10
40	18	0.45	7.7	1.5	0.16	0.043	0.17	0.078	12	3.8
69	7	0.17	0.6	0.2	0.10	0.034	0.06	0.123	12	3.4
83–23	23–15	0.53–0.41	8.1–7.3	1.6–1.4	0.25–0.10	0.071–0.027	0.21–0.14	0.15–0.042	21–7.1	8.0–2.2

and mafic and intermediate igneous rocks belonging to the lower section of the volcano-sedimentary sequence in the >250-m-thick Palma Unit (see Fig. 3 in Farfán et al., 2019). Stringer and stockwork mineralization are also documented by these authors. Farfán et al. (2019) describe that the mineralogy of the upper semi-massive sulfide bodies comprises mainly pyrite, sphalerite, galena, and barite, with subordinate pyrrhotite and chalcopyrite, while the proportion of these two minerals increases in lower massive sulfide bodies. In addition, minor amounts of arsenopyrite and marcasite were identified during this survey. The samples analyzed in this work come from the upper semi-massive sulfide bodies. Representative microphotographs of samples containing the analyzed sphalerite from Palma are presented in Fig. S2. Ore and host rock have partly undergone contact metamorphism induced by arc plutons (Farfán et al., 2019).

2.2. Geology of the Cerro de Maimón deposit, Dominican Republic

The Cerro de Maimón VMS deposit is located 70 km northwest of Santo Domingo in central Dominican Republic (Fig. 3). It stands as the only VMS deposit currently under production in the Caribbean region. It is owned and operated by Perilya Limited through its subsidiary Corporación Minera Dominicana (CORMIDOM). The Cerro de Maimón massive sulfide body contains 4.8 Mt of proven and probable reserves grading 2.54% copper, 0.96 g/t Au, and 34.9 g/t Ag (Perilya, 2020). VMS mineralization and the metallogenic evolution of the Caribbean Greater Antilles are summarized and discussed by Nelson et al. (2011) and Torró et al. (2018a). The Cerro de Maimón massive sulfide mineralization is hosted by the Aptian Maimón Formation (Fig. 3), which is composed mostly of bimodal mafic-felsic volcanic and volcanoclastic rocks of boninitic and tholeiitic affinities emplaced in the forearc portion of the nascent Caribbean island-arc (Torró et al., 2016a, 2017, 2018b). Massive sulfides and hosting volcanic rocks were deformed and metamorphosed to peak metamorphic conditions in the greenschist/blueschist facies transition (Torró et al., 2016b). The mineralization is composed of pyrite grains with both cataclastic/blow-apart and metablastic welding textures and an interstitial assemblage of chalcopyrite + sphalerite + quartz + tennantite ± galena (paragenetic sequence and photomicrographs available in Fig. 6 in Torró et al., 2016a).

3. Materials and methods

The study is based on 36 ore samples from the María Teresa (n = 24), Palma (n = 4), and Perubar (n = 5) deposits in Peru and the Cerro de

Maimón deposit (n = 3) in the Dominican Republic. A list of the samples, their location, and brief textural descriptions are given in Table S1 in the Supplementary material. In the María Teresa deposit, 15 ore samples picked up at different depths along a single drill core (34-18-S) that cut the zone-refined Sofia-D sulfide body (Calera sector) have been used to monitor mineralogical and mineral geochemical variations with depth. A total of 34 thick polished sections were produced at the QEMSCAN laboratory of the Pontificia Universidad Católica del Perú (PUCP) and examined using petrographic and scanning electron (SEM) microscopy. SEM images and major element mineral compositions were acquired on an environmental SEM Thermo Fisher Quanta 650 FEI equipment with an EDAX-Octane Pro EDS microanalysis system at PUCP (Centro de Caracterización de Materiales – CAM-PUCP) and a VEGA II XMU Tescan equipment with a Bruker QUANTAX EDS microanalysis system at BIZALAB at operating conditions of 5 nA and 20 kV accelerating voltage.

Minor and trace element compositions of sphalerite and chalcopyrite were determined by means of electron microprobe (EPMA), at the University of Barcelona, and laser-ablation inductively-coupled plasma mass spectrometry (LA-ICP-MS), at the ETH Zürich, using the same equipment and methodology as described in Benites et al. (2021a). For samples from the María Teresa, Palma, and Perubar deposits, representative EPMA analyses of sphalerite and chalcopyrite are displayed in Table S2 in Supplementary material. For samples from the Cerro de Maimón deposit, representative EPMA analyses of sulfide minerals are provided in Torró et al. (2016a). LA-ICPMS data are reported in Table S3 in Supplementary material. Content statistical data for sphalerite and chalcopyrite are shown in Tables 1–2.

4. Results

4.1. Ore mineral geochemistry

Major and trace element compositions of sphalerite and chalcopyrite show, in general, a wide range of elements at measurable contents. Summaries of their compositions, including minimum, maximum, geometric mean, standard deviation, and interquartile range (IQR) values are displayed in Tables 1 and 2. Content values will be hereinafter reported as IQR unless otherwise specified. Significant effort was made to report only the content of elements whose variations respond to solid solutions and not to mixed mineral analyses by analyzing mineral volumes free of obvious inclusions and by selecting only stable signal intervals in LA-ICP-MS spectra (Fig. S3 in Supplementary material). However, it should be noted that evenly distributed nano-inclusions may

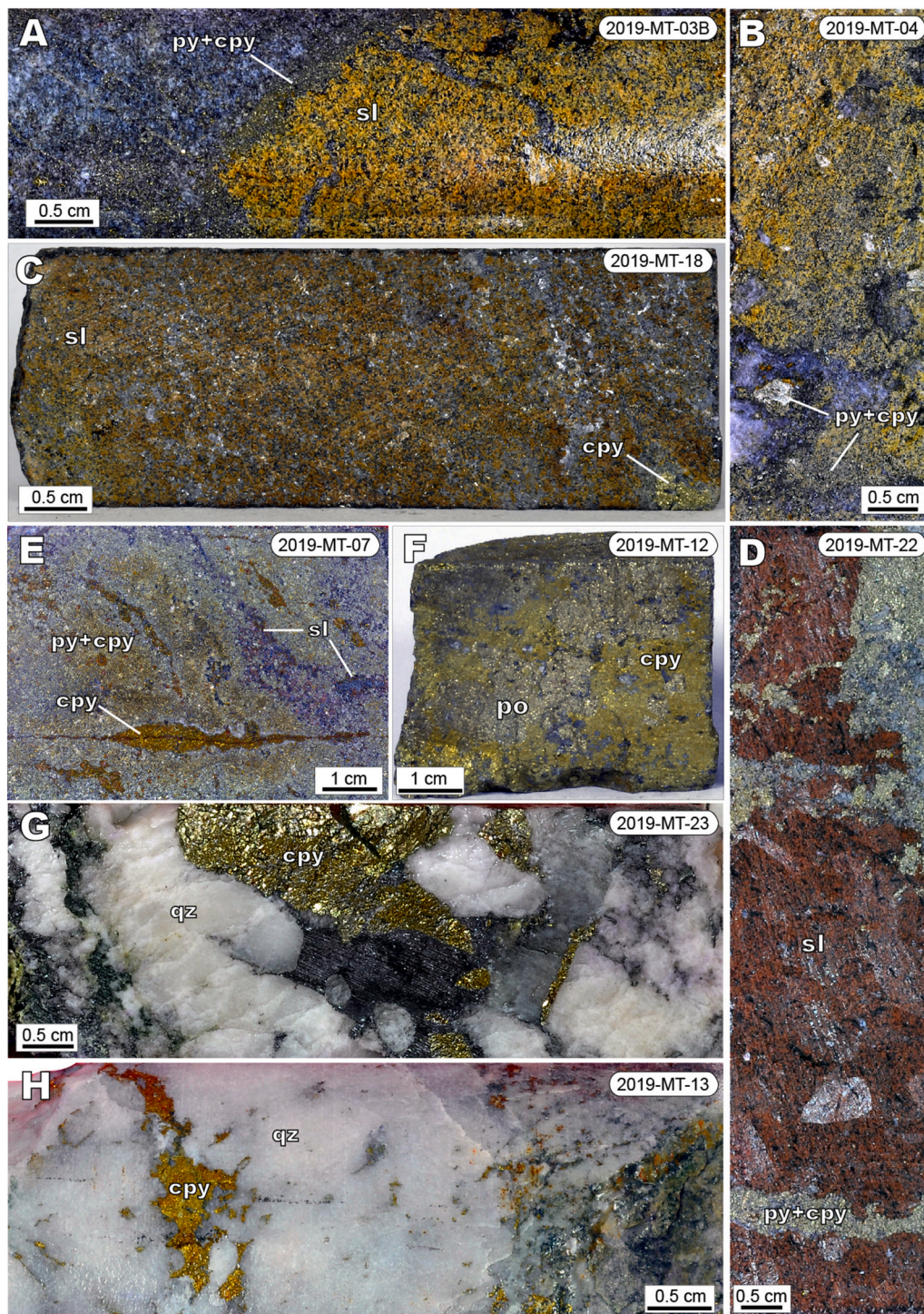


Fig. 4. Drill core photographs of sulfide mineralization from the Sofia-D massive sulfide body in the María Teresa VMS deposit. (A–C) Semi-massive sulfide ore dominated by medium-grained pale-brown sphalerite grains partially replaced by an assemblage of pyrite and chalcopyrite. (D) Massive sulfide ore composed of coarse-grained red-brownish sphalerite grains cut by irregular veinlets of chalcopyrite and pyrite. (E) Dark red, medium-sized sphalerite partially replaced by an assemblage of pyrite and chalcopyrite and the whole assemblage cut by thin, irregular chalcopyrite veinlets. (F) Coarse-grained pyrrhotite and chalcopyrite in a sample picked up close to a contact with a porphyritic andesitic dyke. (G) Silicified volcanoclastic rock with chalcopyrite mineralization along clast contacts. (H) Irregular veinlets of chalcopyrite in a silicified intrusive rock. Key: cpy = chalcopyrite; po = pyrrhotite; py = pyrite; qz = quartz; sl = sphalerite. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

also yield flat ablation profiles (e.g., Gregory et al., 2015) and that, therefore, the possibility of mixed mineral analyses cannot be completely discarded due to the relatively low spatial resolution of the used methods.

4.1.1. Sphalerite

Iron, Mn, Cd, Hg, and Cu (Fig. S3 in Supplementary material) yield smooth and constant ablation profiles. Studied sphalerite shows a relatively wide range of Fe contents in samples from the María Teresa deposit (40675–17,122 ppm), which are higher than in samples from the Cerro de Maimón (585–535 ppm) and generally lower than in samples

from the Perubar (67570–39,965 ppm) and Palma (63423–30,015 ppm) deposits (Table 1). Manganese also shows a relatively wide range of contents in sphalerite from María Teresa (2343–326 ppm), which overlaps with the composition of grains from Cerro de Maimón (482–446 ppm) and are in general lower than in grains from Perubar (5964–907 ppm) and Palma (7751–6734 ppm). The contents of Cd are constantly high in all analyzed sphalerite samples (7373–5290 ppm in María Teresa, 6665–4297 ppm in Perubar, 7109–4782 ppm in Palma, and 3829–3574 ppm in Cerro de Maimón). Mercury contents are higher in sphalerite from the Palma (667–593 ppm) and María Teresa (448–326 ppm) deposits than in sphalerite from the Perubar (107–18

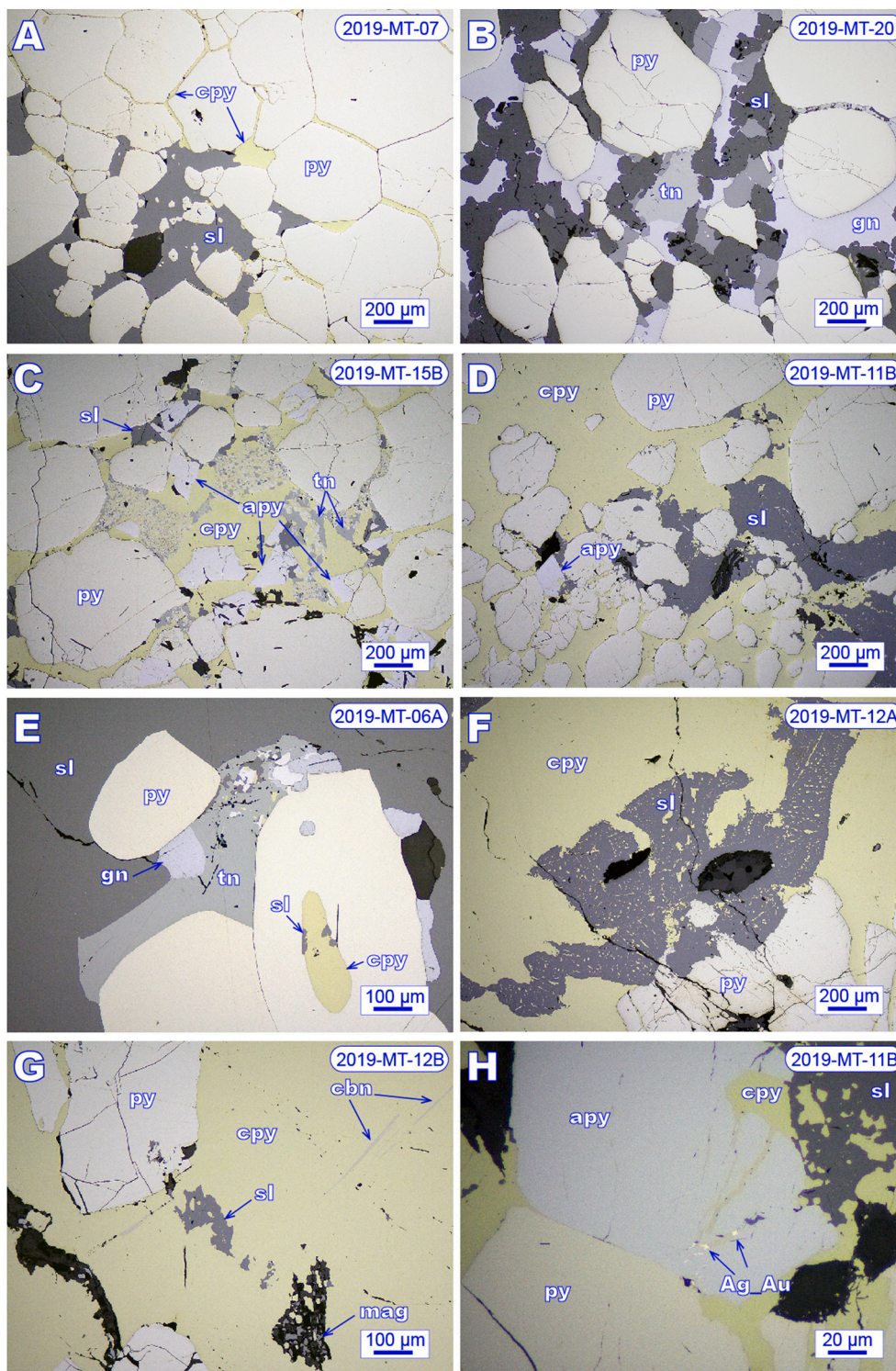


Fig. 5. Photomicrographs (reflected light) of textural features in the samples from the Sofia-D massive sulfide bodies in the María Teresa deposit used for this study. (A) Foam textured pyrite with triple junctions at 120° along with interstitial chalcopyrite and sphalerite. (B) Anhedral roundish and sub-roundish pyrite grains with irregular fractures are surrounded by an assemblage of sphalerite, galena and tennantite. (C–D) Anhedral, sub-roundish pyrite and subhedral arsenopyrite grains both with internal fractures with interstitial chalcopyrite, sphalerite, and tennantite. (E) Detail of a sphalerite-galena-tennantite intergrowth between roundish pyrite grains; note the general foam texture described by the contacts between all phases. (F) Pervasive replacement of sphalerite by chalcopyrite along borders and as fine disseminations (chalcopyrite disease texture). (G) Detail of chalcopyrite with exsolutions of sharply bounded cubanite laths. (H) Detail of minute inclusions of auriferous silver in chalcopyrite veinlets crosscutting arsenopyrite. Key: apy = arsenopyrite; Ag Au = auriferous silver; cbn = cubanite; cpy = chalcopyrite; mag = magnetite; po = pyrrhotite; py = pyrite; qz = quartz; sl = sphalerite; tn = tennantite.

ppm) and Cerro de Maimón (26–22 ppm) deposits. The content of Cu is generally higher in sphalerite samples from the Cerro de Maimón (138–113 ppm) and María Teresa (98–36 ppm) deposits than in sphalerite samples from the Perubar (41–25 ppm) and Palma (31–11 ppm) deposits.

At generally lower contents and also showing smooth ablation spectra are Ag, In, Ge, and Ga (Table 1; Fig. S3 in Supplementary material). Silver content ranges mostly overlap in sphalerite grains from the four studied deposits (12–4.2 ppm in María Teresa, 7.1–3.6 ppm in Perubar, 17–3.8 ppm in Palma, and 4.2–3.1 ppm in Cerro de Maimón).

The contents of In are in general much higher in sphalerite from the María Teresa deposit (68–4.4 ppm, up to 415 ppm) than in sphalerite from the Cerro Maimón (10–9.5 ppm), Perubar (2.3–0.78 ppm) and Palma (0.22–0.0091 ppm; Table 1) deposits. Gallium contents are higher in sphalerite from the Cerro de Maimón deposit (95–78 ppm, up to 96 ppm) than in sphalerite from the María Teresa (32–11 ppm, up to 94 ppm), Perubar (30–4.7 ppm), and Palma (2.4–1.9 ppm) deposits. Germanium contents are in general very low in analyzed sphalerite grains (0.65–0.17 ppm in María Teresa, 0.19–0.15 ppm in Perubar, 3.5–0.16 ppm in Palma, and 1.2–0.68 ppm in Cerro de Maimón).

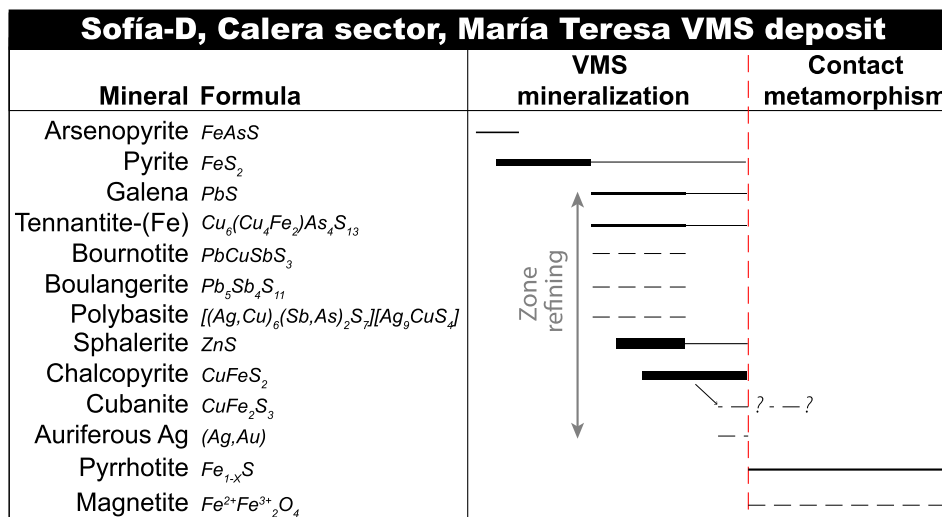


Fig. 6. Paragenetic sequence for the Sofía-D sulfide body in the Calera area, María Teresa VMS deposit. Thick bars indicate higher abundances and dashed lines, lesser abundances.

The content of other elements are in general very low (Pb: 9.3–1.1 ppm; Sb: 3.7–0.31 ppm; Tl: 0.66–0.045; Sn: 0.49–0.23 ppm; Bi: 0.60–0.0055 ppm; detailed results for individual deposits are available in Table 1). Some analyses show slightly ragged ablation spectra (Fig. S1 in Supplementary material).

4.1.2. Chalcopyrite

Chalcopyrite grains from the María Teresa and Cerro de Maimón VMS deposits have been analyzed. Zinc, Ag, Sn, Cd, Se, In, Ge, and Ga show relatively constant ablation profiles (Fig. S3 in Supplementary material). Zinc is the trace element most highly concentrated in studied sphalerite grains; its contents are higher in chalcopyrite from the María Teresa deposit (425–194 ppm) than in chalcopyrite from the Cerro de Maimón deposit (a single analysis >d.l. yields a concentration of 71

ppm; Table 2). Silver, Sn, and Cd are relatively enriched in chalcopyrite grains from the María Teresa deposit (150–71 ppm Ag, 77–33 ppm Sn, 12–6.2 ppm Cd) relative to grains from the Cerro de Maimón deposit (23–15 ppm Ag, 1.6–1.4 ppm Sn, 0.53–0.41 ppm Cd). Selenium is more concentrated in chalcopyrite from the Cerro de Maimón deposit (89–79 ppm) than in chalcopyrite from the María Teresa deposit (31–19 ppm). Indium values are relatively constant in analyzed grains from the studied deposits and are systematically higher in chalcopyrite from María Teresa (32–12 ppm) than in chalcopyrite from Cerro de Maimón (8.1–7.3 ppm). The contents of Ga and Ge are in general lower than those of In (9.2–2.6 ppm Ga and 4.3–0.66 ppm Ge in María Teresa; 3.6–3.3 ppm Ga and 6.1–5.3 ppm Ge in Cerro de Maimón).

Downhole ablation spectra reveal very irregular profiles for Mo and Au and in some samples, slightly irregular profiles for Tl, Sb, and Bi (Fig.

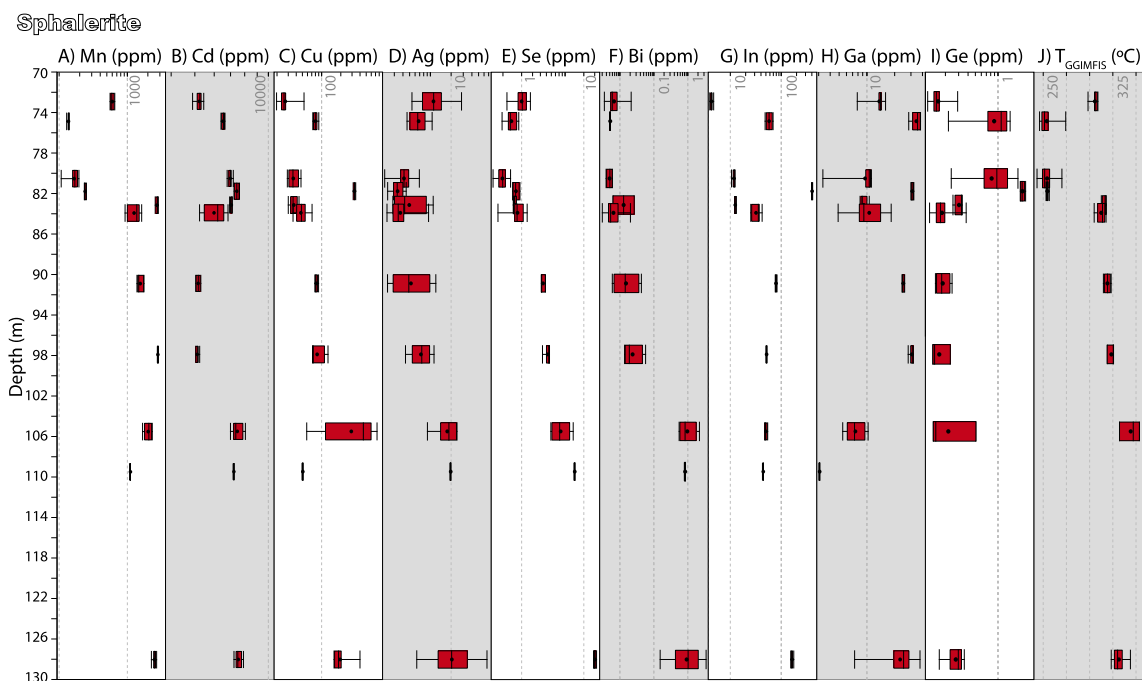
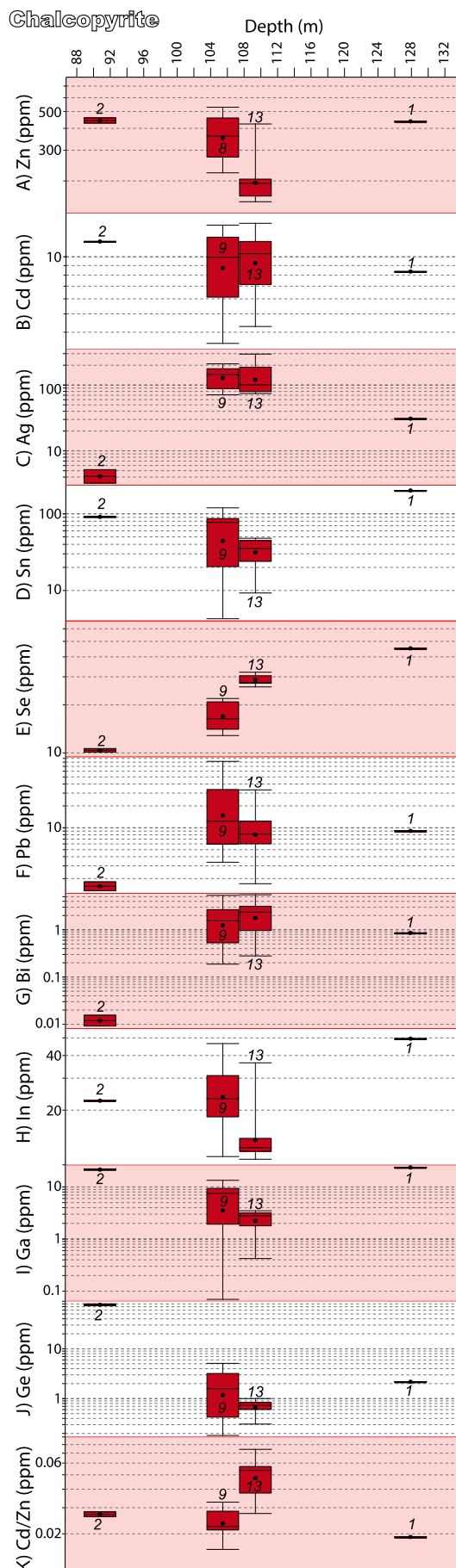


Fig. 7. Boxplots of log-transformed element contents (A-I) and crystallization temperature (J) according to the GGIMFIS geothermometer of Frenzel et al. (2016) for sphalerite at different depths along the drill core 34-18-S in the mine section 300 across the Sofía-D massive sulfide body, María Teresa deposit (LA-ICP-MS data). Box-and-whisker plots provide IQR (box), 1.5 IQR (whiskers), median (line), and outliers (circles).



(caption on next column)

Fig. 8. Boxplots of log-transformed element contents in chalcopyrite at different depths along the drill core 34-18-S in the mine section 300 across the Sofía-D massive sulfide body, Calera sector, María Teresa deposit (LA-ICP-MS data). Box-and-whisker plots provide IQR (box), 1.5 IQR (whiskers), median (line), and outliers (circles).

S3 in Supplementary material). Molybdenum yields relatively erratic contents, which are much higher in analyzed chalcopyrite from Cerro de Maimón (83–23 ppm) than in chalcopyrite from María Teresa (0.034–0.021 ppm, and mostly <d.l.). The contents of Tl (<11 ppm), Sb (<31 ppm), Bi (<10 ppm), and Au (<0.40 ppm) are relatively low (Table 2).

4.2. Metal zonation, mineralogy and trace-element distribution patterns in the zone-refined Sofía-D body

The metal zonation in Sofía-D massive sulfide body includes, bottom to top, i) a Cu zone composed dominantly of chalcopyrite and lesser amounts of sphalerite grading downward to massive pyrite; ii) a Zn zone in which sphalerite is the most abundant phase and appears along with lesser amounts of galena and chalcopyrite and minor fahlore; and iii) a Pb – Zn – Ag zone with abundant galena and sphalerite, and lesser proportions of fahlore and barite (Fig. 2). In detail, this zoning is also recognized from the steep feeders outwards.

On the hand sample scale, sphalerite occurs as mm- to cm-sized grains of yellow-pale brown (Fig. 4A–C) and red-brownish (Fig. 4D–E) color. In places, the massive sphalerite aggregates are cut by irregular, chalcopyrite-rich veinlets (Fig. 4A, D–E). In the eastern part of the María Teresa deposit, i.e., close to the granodiorite stock as is the case for the Sofía D ore body, coarse-grained pyrrhotite ± magnetite is observed and, similarly to in Perubar (see Vidal, 1987; Polliand et al., 1999), interpreted as the metamorphic product after pyrite. An example is shown in Fig. 4F coming from a 20-cm-wide band composed of coarse-grained pyrrhotite ± magnetite at the contact with and andesitic dyke. In volcanoclastic rocks in the immediate host of the Sofía-D body, chalcopyrite selectively replaced some of the clasts and the matrix (Fig. 4G). Silicified dykes that cut the Sofía-D body typically contain patches and irregular veinlets of chalcopyrite (Fig. 4H).

Under the petrographic microscope, pyrite grains, which have sizes typically in the range between 100 and 600 μm , draw a variety of textures including anhedral isolated grains with roundish and embayed outlines, aggregates of several grains often with foam equilibrium textures (i.e., contacts at 120°) and, in some cases, internal irregular fractures (Fig. 5A–D). Arsenopyrite is relatively scarce in the studied samples and appears as isolated euhedral and subhedral fractured grains and sizes in the range between 50 and 200 μm (Fig. 5C–D). The other sulfides, fahlore (tennantite-(Fe) ± argentian tetrahedrite), and traces of other sulfosalts such as bournonite, boulangerite, and polybasite (petrographic observations complemented with EDS analyses) occupy interstitial space between, and line porosity and fractures within, pyrite ± arsenopyrite grains (Fig. 5). According to these observations, pyrite and arsenopyrite crystallized early in the paragenetic sequence (Fig. 6) and, in places, underwent thermal re-equilibration connected to dyke and granodiorite intrusions (Pichardo et al., 2019). Internal fractures of pyrite and arsenopyrite (Fig. 5C) also point to fragile deformation, probably due to movements along the Sofía reverse fault (Fig. 2). Sphalerite grains in the Zn zone show clean surfaces (Fig. 5B,E), whereas in the Cu zone they have been pervasively replaced by chalcopyrite and show dense chalcopyrite disease textures (Fig. 5D, F–G). Chalcopyrite floods the interstitial space between pyrite ± arsenopyrite in the Cu zone along with relicts of sphalerite and in places shows sharply bounded exsolutions of cubanite laths up to 150 μm in length (Fig. 5G). Auriferous silver (73 wt% Ag, 27 wt% Au, EDS analysis) has been observed in one sample and appears as minute grains (<2 μm) in fractured arsenopyrite along with chalcopyrite (Fig. 5H).

Table 3

Summary of LA-ICP-MS data for sphalerite from subduction-related VMS deposits reported in the literature.

Deposit}	Location}	ppm}	Mn}	Fe}	Cu}	Ga}	Ge}	As}	Se}
Vorta (n = 8)	Romania	Max	761	2617	719	363	23	n.a.	1.4
		Min	165	917	131	75	2.7	n.a.	0.82
		GM	485	1844	334	118	8.0	n.a.	1.1
		SD	201	584	193	100	7.2	n.a.	0.18
		IQR	650–424	2404–1524	475–274	171–79	12–5.8	n.a.	1.2–1.0
Eskay Creek (n = 12)	Canada	Max	4744	7112	3483	857	14	1550	4.1
		Min	2296	3880	67	19	0.47	1	1.3
		GM	3206	4814	356	86	1.2	27	2.1
		SD	648	900	1017	228	3.7	435	1.3
		IQR	3437–3113	5284–4316	835–116	123–61	1.4–0.89	196–3.9	3.3–1.5
Zinkgruvan (n = 5)	Sweden	Max	532	33,261	5.1	2.8	1.2	n.a.	1.9
		Min	499	31,735	1.9	1.9	0.82	n.a.	1.9
		GM	513	32,330	3.3	2.2	1.1	n.a.	1.9
		SD	14	737	1.2	0.38	0.15	n.a.	–
		IQR	524–502	33,006–31,751	4.0–3.1	2.4–1.9	1.2–1.1	n.a.	1.9–1.9
Marketorp (n = 8)	Sweden	Max	18,729	119,418	14,368	9.6	13	n.a.	36
		Min	1684	86,959	1414	0.61	1.8	n.a.	29
		GM	2606	96,260	3020	1.5	2.7	n.a.	33
		SD	5930	12,125	4246	3.1	3.7	n.a.	1.8
		IQR	2373–1714	100,257–88,877	3290–2281	1.8–0.87	2.6–1.9	n.a.	34–33
Sauda (n = 10)	Norway	Max	1449	120,205	44	8.6	1.4	n.a.	109
		Min	874	69,576	22	4.1	0.86	n.a.	89
		GM	959	77,585	28	5.5	1.0	n.a.	94
		SD	170	14,797	8.6	1.4	0.16	n.a.	6.0
		IQR	939–903	75,796–72,800	37–23	6.0–4.7	1.1–0.92	n.a.	96–90
Çayeli (n = 40)	NE Turkey	Max	486	182,759	210,618	n.a.	n.a.	7261	62
		Min	0.35	8.0	65	n.a.	n.a.	0.19	0.13
		GM	22	2857	9638	n.a.	n.a.	345	7.1
		SD	102	32,933	41,584	n.a.	n.a.	1874	12
		IQR	83–5.2	16,904–822	35,768–6982	n.a.	n.a.	1693–275	13–5.0
Laochang (n = 30)	S China	Max	4111	154,203	2020	117	15	36	4.0
		Min	2626	121,694	119	2.3	2.1	0.33	0.62
		GM	3040	130,599	211	15	3.8	2.2	1.6
		SD	364	7814	339	24	2.5	9.7	1.1
		IQR	3255–2724	137,135–125,803	254–151	24–8.9	3.9–2.9	4.1–0.93	2.5–1.0

B.D.L.: Below detection limit; n.a.: not analyzed; GM: geometric mean; SD: standard deviation; IQR: interquartile range.

The systematic sampling developed along the drill core 34-18-S from 72.85 to 128 m (Table S1 in the Supplementary material) allows for the assessment of vertical trends in trace-element contents in sphalerite and chalcopyrite. Bottom to top in sphalerite, there is a relatively sustained decrease in the contents of Mn, Cu, Se, Bi, and In and an increase of Ge, and the contents of elements such as Cd and Ga show irregular distributions (Fig. 7A–G). The contents of In significantly decrease from sample 2019-MT-015 (128 m, 167–158 ppm) to sample 2019-MT-001 (72.85 m, 4.4–4.1 ppm), although the maximum contents occur in sample 2019-MT-004 (81.4 m, 409–399 ppm; Fig. 7G). The distribution of Cu contents almost mimics that of In, with a decrease from sample 2019-MT-015 (128 m, 203–160 ppm) to sample 2019-MT-001 (72.85 m, 27–24 ppm), although with maximum contents at intermediate depths in samples 2019-MT-004 (81.4 m, 335–321 ppm) and 2019-MT-011 (105.5 m, 597–115 ppm; Fig. 7C). Germanium contents show a moderately irregular distribution pattern with a general increase in its content from sample 2019-MT-015 (128 m, 0.33–0.23 ppm) to sample 2019-MT-002 (74.6 m, 1.3–0.73 ppm; Fig. 7I). As for chalcopyrite, vertical distribution patterns are mostly irregular (Fig. 8). The most remarkable exception is Se, which content tends to decrease steadily at shallower levels (45 ppm – single value – in sample 2019-MT-015 at 128 m, 31–27 ppm in sample 2019-MT-012 at 109.4 m, 11–10 ppm in sample 2019-MT-007 at 90.5 m; Fig. 8E). An opposite trend is defined by the content of Ge, with general enrichment from deeper (2.2 ppm – single value – in sample 2019-MT-015 at 128 m, 0.84–0.60 ppm in sample 2019-MT-012 at 109.4 m,) to shallower (80–74 ppm in sample 2019-MT-007 at 90.5 m) chalcopyrite samples (Fig. 8J). Although probably hosted as sub-microscopic mineral inclusions, the content of Bi in chalcopyrite also shows significant variation and tends to increase with depth (Fig. 8G).

5. Discussion

5.1. Trace-element budget in sphalerite and chalcopyrite from arc-hosted VMS deposits

The minor and trace element contents measured in sphalerite and chalcopyrite from the Palma, Perubar, María Teresa (Perú), and Cerro de Maimón (Dominican Republic) VMS deposits will be compared along the following lines with available data from other VMS deposits, and particularly with those formed in arc systems. We have compared our results with the comprehensive compilation by Monecke et al. (2014) of whole-rock and mineral geochemical data from both active seafloor massive sulfide (SMS) and fossil equivalent deposits emplaced in a variety of tectonic settings worldwide.

Sphalerite from the four investigated deposits yields fairly variable Fe contents (Table 1), the content of the other analyzed trace elements being largely subordinated to that of Fe. The content of Fe in sphalerite samples from the Cerro de Maimón deposit analyzed in this study is particularly low (Table 1); a more complete geochemical dataset based on EPMA analyses on sphalerite samples from this deposit reported by Torró et al. (2016a) indicates average Fe contents of 1.16 wt%. Iron contents in the studied deposits are compatible with sphalerite crystallized in intermediate to high-intermediate sulfidation environments in equilibrium with pyrite (Barton and Toulmin, 1966; Hannington and Scott, 1989). Co-crystallization of fahlore (tennantite and tetrahedrite) along with sphalerite in the studied ore samples is a further indication of intermediate-sulfidation state of the hydrothermal mineralizing fluids. Similar Fe contents to those in the studied Andean VMS deposits are reported in sphalerite samples from the Zinkgruvan, Marketorp, and Sauda VMS deposit (Table 3 and references therein). Iron contents

Mo}	Ag}	Cd}	In}	Sn}	Sb}	Hg}	Tl}	Pb}	Bi}	Reference}
149	21	5088	0.060	0.24	17	n.a.	1.1	3426	0.71	Cook et al. (2009)
8.0	1.4	4169	0.040	0.050	0.040	n.a.	0.010	0.46	0.020	
40	3.7	4734	0.049	0.080	1.1	n.a.	0.034	12	0.16	
50	6.7	337	0.014	0.072	5.7	n.a.	0.50	1201	0.27	
71–20	5.3–2.0	5044–4562	0.055–0.045	0.080–0.055	2.3–0.75	n.a.	0.040–0.010	68–2.1	0.49–0.085	Cook et al. (2009)
31	2785	4955	14	31	117,467	n.a.	21	50,955	0.19	
0.15	6.8	2720	0.10	2.6	5.1	n.a.	0.081	1.2	0.020	
0.63	102	3796	0.29	9.1	1034	n.a.	0.83	467	0.041	
8.7	872	679	3.9	11	33,692	n.a.	7.1	15,639	0.056	Cook et al. (2009)
0.77–0.29	575–15	4394–3294	0.45–0.10	19.8–5.2	31,973–33	n.a.	2.3–0.17	17,228–13	0.053–0.028	
n.a.	13	3943	0.32	0.25	4.1	n.a.	n.a.	193	0.15	
n.a.	4.6	3661	0.26	0.080	0.12	n.a.	n.a.	1.4	0.060	
n.a.	6.0	3811	0.29	0.13	0.54	n.a.	n.a.	8.0	0.094	Cook et al. (2009)
n.a.	3.5	126	0.026	0.077	1.7	n.a.	n.a.	84	0.033	
n.a.	6.1–4.6	3933–3718	0.31–0.27	0.21–0.090	1.4–0.24	n.a.	n.a.	15–2.8	0.10–0.080	
0.68	34	3619	28	0.81	0.080	n.a.	0.30	29	74	
0.11	7.4	2968	25	0.090	0.040	n.a.	0.024	0.81	2.2	Cook et al. (2009)
0.27	12	3246	26	0.20	0.057	n.a.	0.072	7.1	13	
0.40	8.8	191	1.1	0.26	0.019	n.a.	0.10	9.6	24	
0.54–0.25	14–8.7	3304–3135	27–26	0.29–0.13	0.070–0.040	n.a.	0.11–0.045	17–6.5	30–4.3	
0.17	5.5	3488	42	1.3	0.10	n.a.	n.a.	n.a.	0.99	Cook et al. (2009)
0.17	2.4	1973	27	0.27	0.010	n.a.	n.a.	n.a.	0.12	
0.17	3.7	2170	29	0.56	0.018	n.a.	n.a.	n.a.	0.31	
–	1.0	455	4.4	0.38	0.035	n.a.	n.a.	n.a.	0.27	
0.17–0.17	4.5–3.1	2117–2031	30–28	0.97–0.38	0.020–0.010	n.a.	n.a.	n.a.	0.47–0.18	Revan et al. (2014)
222	450	8957	n.a.	512	3354	n.a.	97	171,820	254	
0.030	0.80	1051	n.a.	0.43	0.68	n.a.	0.030	8.0	B.D.L.	
5.0	51	3234	n.a.	14	141	n.a.	5.2	727	–	
45	98	1936	n.a.	81	632	n.a.	20	26,976	–	Ye et al. (2011)
18–0.99	138–40	3430–2636	n.a.	44–4.4	683–52	n.a.	13–4.9	2123–549	113–6.8	
n.a.	10	9600	566	38	4.1	n.a.	2.6	556	0.050	
n.a.	4.8	8306	66	2.2	B.D.L.	n.a.	0.0060	0.30	B.D.L.	
n.a.	6.7	8736	184	5.6	–	n.a.	0.19	2.9	–	
n.a.	1.3	240	89	6.9	–	n.a.	0.61	101	–	
n.a.	7.5–5.9	8853–8562	235–164	7.7–3.9	1.6–0.10	n.a.	0.72–0.053	9.2–0.58	0.010–0.010	

similar to those found in sphalerite from the Cerro de Maimón deposit are reported in the Vorta, Eskay Creek, and Çayeli VMS deposits. In contrast, sphalerite from the VMS Laochang deposit (South China) systematically yields higher Fe contents.

The relatively high content of Cd in sphalerite from the studied deposits (Table 1) are comparable with those in sphalerite from other VMS deposits such as Vorta, Eskay Creek, Zinkgruvan, Marketorp, and Laochang, and higher than in Sauda (Table 3). Cadmium contents of several hundreds of parts per million are typical of SMS emplaced in arc settings compared to contents in the range of some tens of parts per million in those emplaced in mid-ocean ridge systems, sphalerite and wurtzite being the main Cd hosts in VMS deposits according to Monecke et al. (2014). According to our (Table 1) and previous (Table 3) data, Cd in sphalerite from VMS deposits in arc systems typically yields contents of several thousands of parts per million. The content of Hg varies between some tens (Perubar and Cerro de Maimón) and some hundreds (María Teresa and Palma) of parts per million in the studied sphalerite samples; the content of Hg is not provided for sphalerite from other VMS deposits in arc systems in the revised literature. Petersen et al. (2014) reported contents between <0.02 and 7.4 wt% Hg (median = 0.37 wt% Hg; EPMA data) in sphalerite from the Palinuro SMS deposit in the Aeolian arc (transitional arc setting). Monecke et al. (2014) found out that the content of Hg in SMS is higher in mineralization emplaced in arc volcanoes and rifted arcs (some tens of ppm) relative to mineralization emplaced in back-arc spreading centers and mid-ocean ridges; also, that Hg usually shows a strong positive correlation with Sb in mineralized samples from SMS which normally results from the chief concentration of Hg in tetrahedrite and tennantite over other ore minerals. The contents of Sb in the studied sphalerite are mostly in the 1–10 ppm range or below (Table 1), similar to sphalerite from other VMS mineralization such as Vorta, Zinkgruvan, and Laochang, and lower than in Eskay Creek

and Çayeli (Table 3). The content of Tl in the studied sphalerite samples is normally below 1 ppm (Table 1), similar to sphalerite from other VMS deposits with the exception of Çayeli, in which the content of Tl is much higher (13–4.9 ppm; Table 3). The contents of Sb and Tl are usually higher in sulfide samples from SMS mineralization emplaced in arc volcanoes and rifted arcs than in SMS mineralization emplaced in back-arc spreading centers and mid-ocean ridges according to data compiled by Monecke et al. (2014).

The studied sphalerite samples yield In contents largely in the range between <1 ppm and some tens of parts per million (Table 1), similar to sphalerite from Vorta, Eskay Creek, Zinkgruvan, Marketorp, and Sauda, and lower than in Laochang (Table 3). The measured In contents are also much lower than in sphalerite from the Neves Corvo VMS deposit (up to 2.1 wt%, EPMA data: Carvalho et al., 2018). The contents of Ga in the sphalerite samples analyzed here are largely of the same order of magnitude (from a few ppm to ~100 ppm) as contents reported for sphalerite from other VMS deposits (Table 3), whereas the contents of Ge (mostly <1 ppm) are generally lower than those reported in the literature. The contents of In, Ge, and Ga are mostly indistinguishably low in SMS sulfides irrespective of their tectonic environment of emplacement (Monecke et al., 2014).

In the studied chalcopyrite samples from the Andean VMS deposits, the content of Zn is of some hundreds of parts per million (Table 2), similar to chalcopyrite from Vorta, Sulitjelma, and Kutlular, and lower than in chalcopyrite from Çayeli (Table 4 and references therein); Zn content in sphalerite from the Cerro de Maimón deposit is significantly lower. The contents of Ag are mostly in the range between a few to some hundreds of parts per million in the analyzed chalcopyrite (Table 2), like chalcopyrite from VMS reported in the literature (Table 4). Selenium contents in the studied chalcopyrite samples are of some tens of parts per million (Table 2), and therefore lower than Se contents in chalcopyrite

Table 4
Summary of LA-ICP-MS data for chalcopyrite from subduction-related VMS deposits reported in the literature.

Deposit	Location	ppm	Co	Ni	Zn	Ga	Ge	As	Se	Mo	Ag	Cd	In	Sn	Sb	Au	Hg	Tl	Pb	Bi	Reference
Vorta (n = 8)	Romania	Min	12	n.a.	199	0.11	n. a.	2.8	n.a.	n.a.	18	0.48	0.0049	0.19	0.11	0.0080	0.34	0.014	6.1	0.010	George et al. (2018)
		Max	12	n.a.	538	2.2	n. a.	5.8	n.a.	n.a.	92	3.2	0.76	0.79	5.6	0.034	2.6	0.17	53	2.2	
		GM	12	n.a.	391	0.71	n. a.	4.1	n.a.	n.a.	45	1.6	0.044	0.39	2.0	0.021	0.62	0.068	14	0.075	
		SD	–	n.a.	130	0.70	n. a.	1.3	n.a.	n.a.	23	1.0	0.43	0.28	1.8	0.012	0.82	0.061	26	1.3	
		IQR	12–12	n.a.	479–395	1.4–0.49	n. a.	5.2–3.2	n.a.	n.a.	55–41	2.2–1.3	0.39–0.014	0.73–0.23	3.8–2.8	0.033–0.019	0.82–0.38	0.14–0.058	30–7.1	1.1–0.015	
Sulițjelma (n = 120)	Norway	Min	0.016	0.049	97	0.017	n. a.	1.2	4.4	n.a.	5.1	0.79	0.301	0.091	0.025	0.0031	0.051	0.0016	0.071	0.019	George et al. (2018)
		Max	28	5.1	861	19	n. a.	5.3	926	n.a.	331	58	26	75	15	0.76	0.74	3.0	28	11	
		GM	0.8	0.42	381	0.43	n. a.	2.4	101	n.a.	71	10	2.1	4.6	0.7	0.032	0.20	0.015	0.93	0.41	
		SD	3.7	1.3	195	4.1	n. a.	1.1	267	n.a.	95	16	6.1	26	3.7	0.14	0.17	0.50	5.4	2.2	
		IQR	2.6–0.27	1.0–0.20	569–329	2.3–0.098	n. a.	3.4–1.7	202–51	n.a.	201–32	29–3.1	6.7–0.61	51–0.43	2.0–0.18	0.088–0.013	0.32–0.12	0.089–0.0036	1.7–0.37	0.89–0.15	
Çayeli (n = 69)	NE Turkey	Min	0.010	B.D.L.	7.0	n.a.	n. a.	B.D.L.	3.6	0.11	4.0	0.040	n.a.	1.1	B.D.L.	B.D.L.	n.a.	0.010	4.0	0.030	Revan et al. (2014)
		Max	166	200	407,977	n.a.	n. a.	5008	408	290	2549	2194	n.a.	96	3524	25	n.a.	243	244,075	1000	
		GM	0.7	–	1305	n.a.	n. a.	–	62	15	46	12	n.a.	15	–	–	n.a.	1.2	307	35	
		SD	27	–	69,136	n.a.	n. a.	–	124	65	375	366	n.a.	24	–	–	n.a.	38	30,588	217	
		IQR	1.8–0.17	7.0–0.38	30,530–56	n.a.	n. a.	656–5.6	225–17	54–4.6	132–12	163–1.3	n.a.	39–8.4	117–2.2	0.64–0.063	n.a.	14–0.14	1813–43	170–7.0	
Kutlular (n = 9)	NE Turkey	Min	B.D.L.	0.010	70	n.a.	n. a.	1.7	220	0.010	1.1	0.67	n.a.	9.1	0.50	0.030	n.a.	0.020	5.0	16	Revan et al. (2014)
		Max	0.90	0.34	5891	n.a.	n. a.	20	340	2223	45	33	n.a.	31	7.8	1.3	n.a.	6.6	494	1764	
		GM	–	0.037	304	n.a.	n. a.	5.4	281	7.7	6.8	2.7	n.a.	18	1.4	0.17	n.a.	0.21	37	93	
		SD	–	0.14	1885	n.a.	n. a.	5.4	46	795	18	10	n.a.	7.1	2.5	0.48	n.a.	2.5	186	587	
		IQR	0.26–0.010	0.21–0.010	751–98	n.a.	n. a.	7.3–2.8	334–258	265–0.20	7.0–4.14	5.6–0.99	n.a.	22–15	1.4–0.82	0.21–0.090	n.a.	0.59–0.030	55–16	94–50	

B.D.L.: Below detection limit; n.a.: not analyzed; GM: geometric mean; SD: standard deviation; IQR: interquartile range.

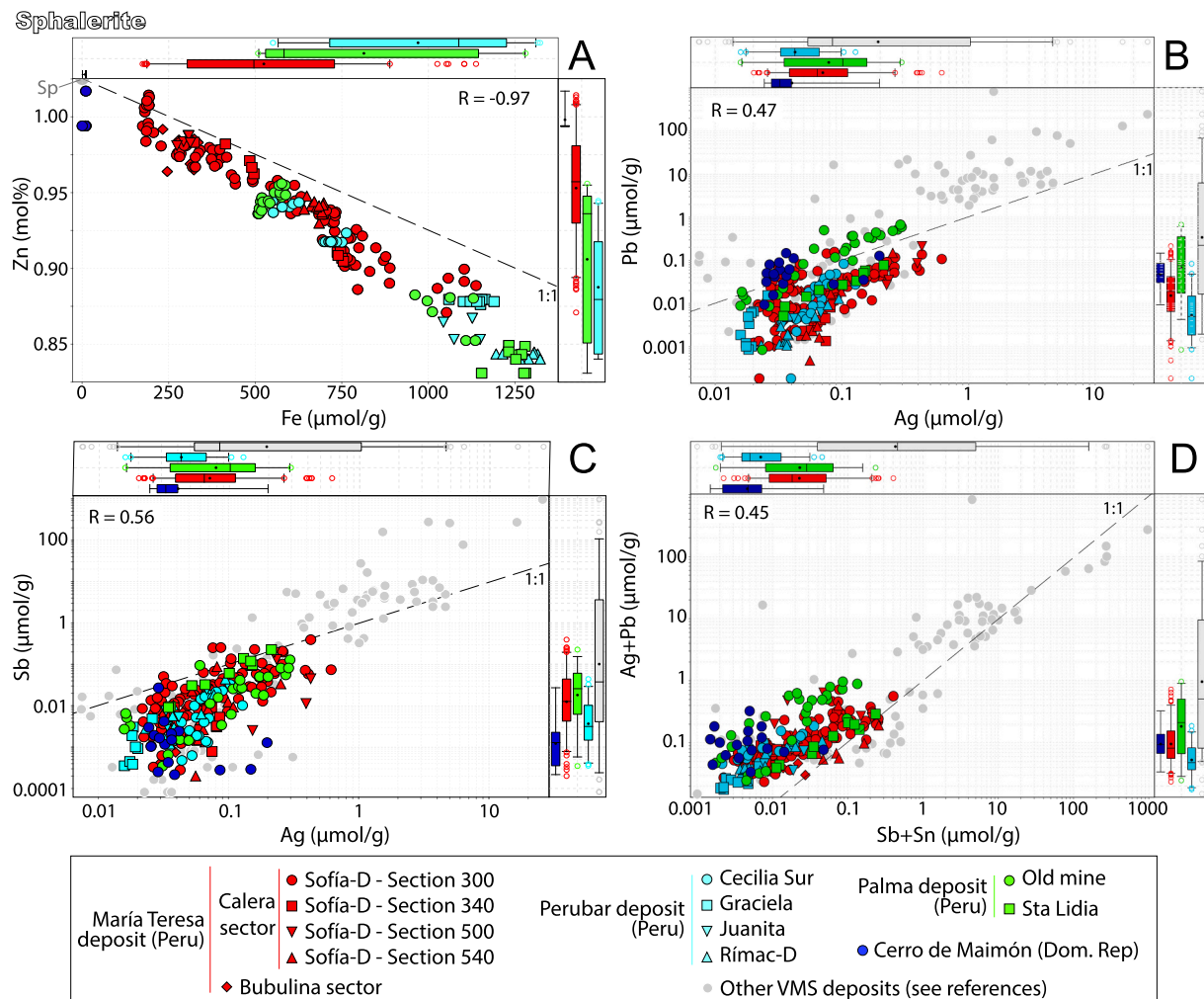


Fig. 9. Binary plots showing inter-element correlations in sphalerite: (A) Fe vs. Zn, (B) Ag vs. Pb, (C) Ag vs. Sb, and (D) Sb+Sn vs. Ag+Pb. Dashed lines represent 1:1 M correlation. Additionally, contents are presented in box-and-whisker plots, which provide IQR (box), 1.5 IQR (whiskers), median (line) and outliers (circles). The composition of sphalerite from other VMS deposits worldwide are shown for comparison (data from [Carvalho et al., 2018](#), [Cook et al., 2009](#), [Revan et al., 2014](#), and [Ye et al., 2011](#)).

from other subduction-related VMS deposits, which are mostly in the hundreds of parts per million range ([Table 4](#)). According to [Monecke et al. \(2014\)](#), chalcopyrite is the main host of Se in VMS and SMS deposits along with pyrite and, typically, higher Se contents are found in sulfide samples collected in SMS formed in mid-ocean ridges relative to those in subduction related systems. Cadmium shows contents of a few to some tens of parts per million in the analyzed chalcopyrite samples ([Table 2](#)), similar to chalcopyrite from other VMS deposits ([Table 4](#)); therefore, as expected, Cd contents are much lower in chalcopyrite than in sphalerite from the same deposit ([Tables 1 and 3](#)). Similarly, Hg contents in chalcopyrite (up to a few ppm, [Table 2](#)) are much lower than in sphalerite ([Table 1](#)) from the respective deposits, and similar to or slightly higher than Hg content in chalcopyrite from the Vorta and Sulitjelma VMS deposits ([Table 4](#)).

Indium content in analyzed chalcopyrite grains, mostly at the level of tens of parts per million, is only slightly lower than In content in sphalerite grains from the same deposits ([Tables 1 and 2](#)), and steadily higher than in chalcopyrite grains from the Vorta and Sulitjelma VMS deposits ([Table 4](#)). Germanium content in the analyzed chalcopyrite samples is mostly at the level of some parts per million ([Table 2](#)) and usually higher than in sphalerite from the respective deposits ([Table 1](#)); Ge content is not provided in reference data for chalcopyrite from other VMS deposits ([Table 4](#)). The content of Ga (mostly at the level of a few ppm to some tens of ppm) is lower in chalcopyrite than in sphalerite

from the same deposit ([Tables 1 and 2](#)); the content of Ga in chalcopyrite from the Vorta and Sulitjelma VMS deposits is systematically lower than in the samples analyzed here ([Table 4](#)).

5.2. Correlation trends and substitution mechanisms

5.2.1. Sphalerite

The dataset shows a marked negative correlation trend between Fe and Zn with a slope of ~ 1 ([Fig. 9A](#)) that indicates a dominant $\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$ simple isovalent substitution. Fair positive correlations between Ag and both Pb ([Fig. 9B](#)) and Sb ([Fig. 9C](#)) suggest that these three elements likely occur in solid solution in the studied sphalerite samples. Tin is also positively correlated with Ag and the reasonable positive correlation in the Sb + Sn vs. Ag + Pb plot ([Fig. 9D](#)) with a slope ~ 1 suggests incorporation of these elements via coupled substitutions of the type $\text{Ag}^+ + \text{Pb}^{2+} + (\text{Sn}, \text{Sb})^{3+} \leftrightarrow 3\text{Zn}^{2+}$ and likely also via independent variations of this scheme ([Cook et al., 2009](#)). Nevertheless, the observation of slight ragged downhole ablation spectra for Sn and Pb in a few analyses (e.g., Fig. S3D in Supplementary material) might suggest the local presence of sub-microscopic inclusions of galena and/or Sn—Pb sulfosalts.

The dataset shows a marked correlation trend between Cu and In in samples from María Teresa, particularly in those with the highest In contents, with a slope ~ 1 ([Fig. 10A](#)) thus pointing to a $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow$

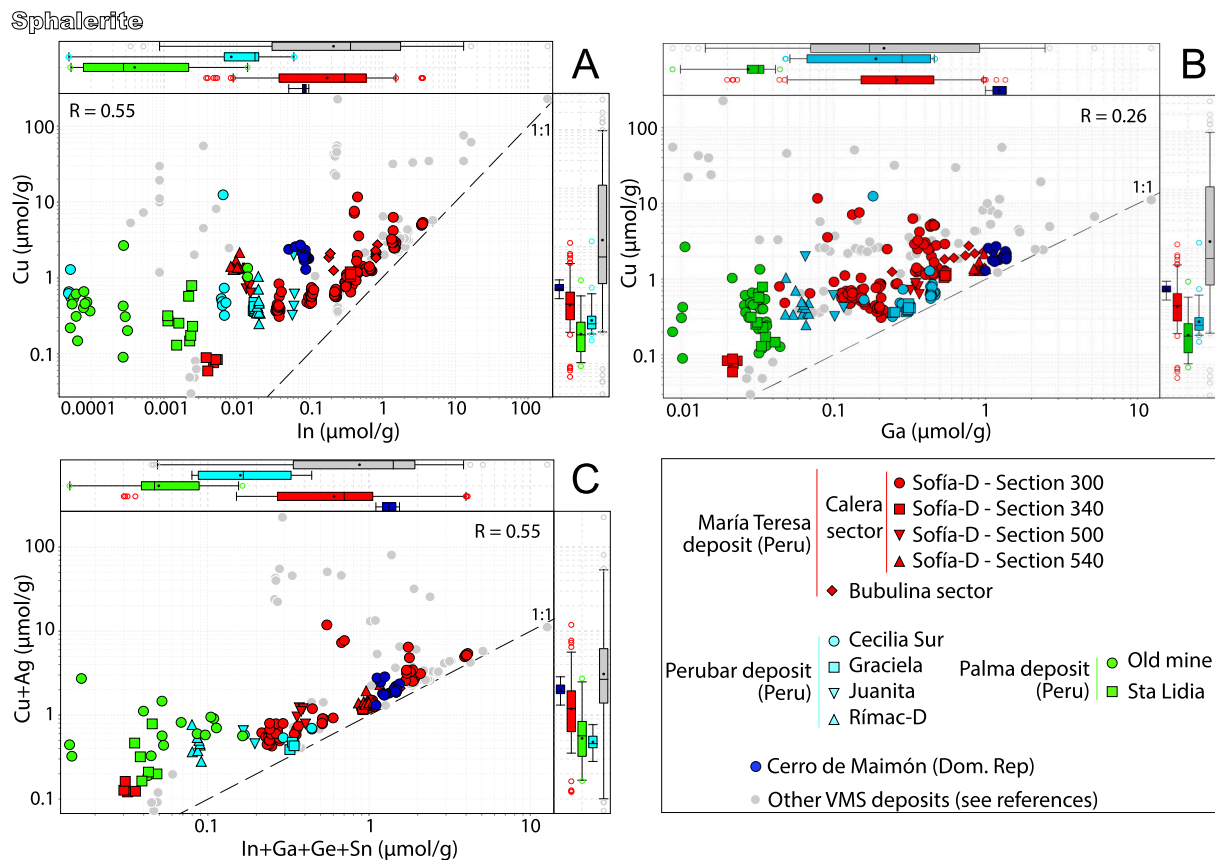


Fig. 10. Binary plots showing inter-element correlations in sphalerite: (A) In vs. Cu, (B) Ga vs. Cu, and (C) Sn+In+Ge+Ga vs. Cu+Ag. Dashed lines represent 1:1 M correlation. Additionally, contents are presented in box-and-whisker plots, which provide IQR (box), 1.5 IQR (whiskers), median (line) and outliers (circles). The composition of sphalerite from other VMS deposits worldwide are shown for comparison (data from [Carvalho et al., 2018](#), [Cook et al., 2009](#), [Revan et al., 2014](#), and [Ye et al., 2011](#)).

2Zn^{2+} coupled substitution ([Cook et al., 2012](#); [Trofimov et al., 2020](#); [Benites et al., 2021a, 2021b](#)). Gallium yields a fair positive correlation with Cu with some analyses plotting along or close to the 1:1 correlation line ([Fig. 10B](#)) suggesting a $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ coupled substitution. Germanium contents do not show clear correlation trends with the other analyzed elements. The alignment of the analyses in the Sn + In+Ge + Ga vs. Cu + Ag plot ($R = 0.55$; [Fig. 10C](#)) would suggest supplementary complex substitution mechanisms involving combinations of coupled substitutions of trivalent cations (i.e., In, Ga) with monovalent cations (i.e., Cu, Ag) and direct substitution of quadrivalent cations (Sn, Ge) with vacancies for Zn. However, the lacking consensus on the oxidation state of Ge (+2 or +4) and Sn (+3 or +4) in sphalerite leaves open the possibility of other substitution schemes (see [Cook et al., 2009, 2015](#); [Bellisont et al., 2014](#)). The low content of both Sn and Ge in the studied sphalerite grains and the apparent lack of individual correlation trends with other analyzed elements do not allow to give a definitive conclusion on their incorporation in the crystal lattice.

5.2.2. Chalcopyrite

Unlike sphalerite, chalcopyrite is not a fully ionic mineral but bonding between atoms in its structure is “effectively” covalent with valencies of Fe fluctuating between divalent and trivalent and valencies of Cu, between monovalent and divalent ([Donnay et al., 1958](#); [Hall and Stewart, 1973](#); [Li et al., 2013](#)). Therefore, the incorporation mechanism of trace elements in chalcopyrite is far more complex to draw than for sphalerite (see [George et al., 2018](#)). Weak to fair positive correlation trends between Zn, Cd, In, Ge, and Ga are observed across our dataset ([Fig. 11](#)). [George et al. \(2018\)](#) also noticed the positive correlation between Zn and Cd in chalcopyrite samples from a variety of

mineralization styles and argued that systematic variations of Cd:Zn ratio are strongly dependent on crystallization temperature (this question is further addressed in the following section). Our data would suggest that the incorporation of Zn, Cd, In, Ge, and Ga to the structure of chalcopyrite is not completely independent (i.e., simple substitution) but that coupled substitutions involving cations of these elements are likely (see also [George et al., 2018](#) and their references).

5.3. Trace element distribution profiles and conditions of crystallization (and re-crystallization)

Metal and mineral zonation ([Fig. 2](#)) and the overprinting and replacement of earlier sulfides (e.g., sphalerite) by minerals that are stable at higher temperature (e.g., chalcopyrite; [Figs. 4–6](#)) as described for Sofia-D in the María Teresa deposit denote a thermally intensifying hydrothermal system during the growth of the massive sulfide body; in VMS systems, this process is commonly known as zone refining ([Eldridge et al., 1983](#); [Hannington et al., 1998](#); [Grant et al., 2015](#); [Almodóvar et al., 2019](#)). According to the GGIMFIS geothermometer of [Frenzel et al. \(2016\)](#) – which uses the contents of Fe, Mn, In, Ga, and Ge in sphalerite, the analyzed sphalerite from the María Teresa deposit crystallized at temperatures in the range between 320 and 257 °C (IQR). Such a temperature range is in good agreement with a VMS mineralization containing fair abundant chalcopyrite ([Hannington et al., 1995](#); [Ohmoto, 1996](#); [Franklin et al., 2005](#)). Assuming that $f\text{O}_2$ and $f\text{S}_2$ were stable during the crystallization of sphalerite (see [Hannington and Scott, 1989](#); [Keith et al., 2014](#)) in the Sofia-D sulfide body, sphalerite composition would record a relatively sustained decrease in the crystallization temperature bottom to top (335–327 °C for sample 2019-MT-015 at 128 m

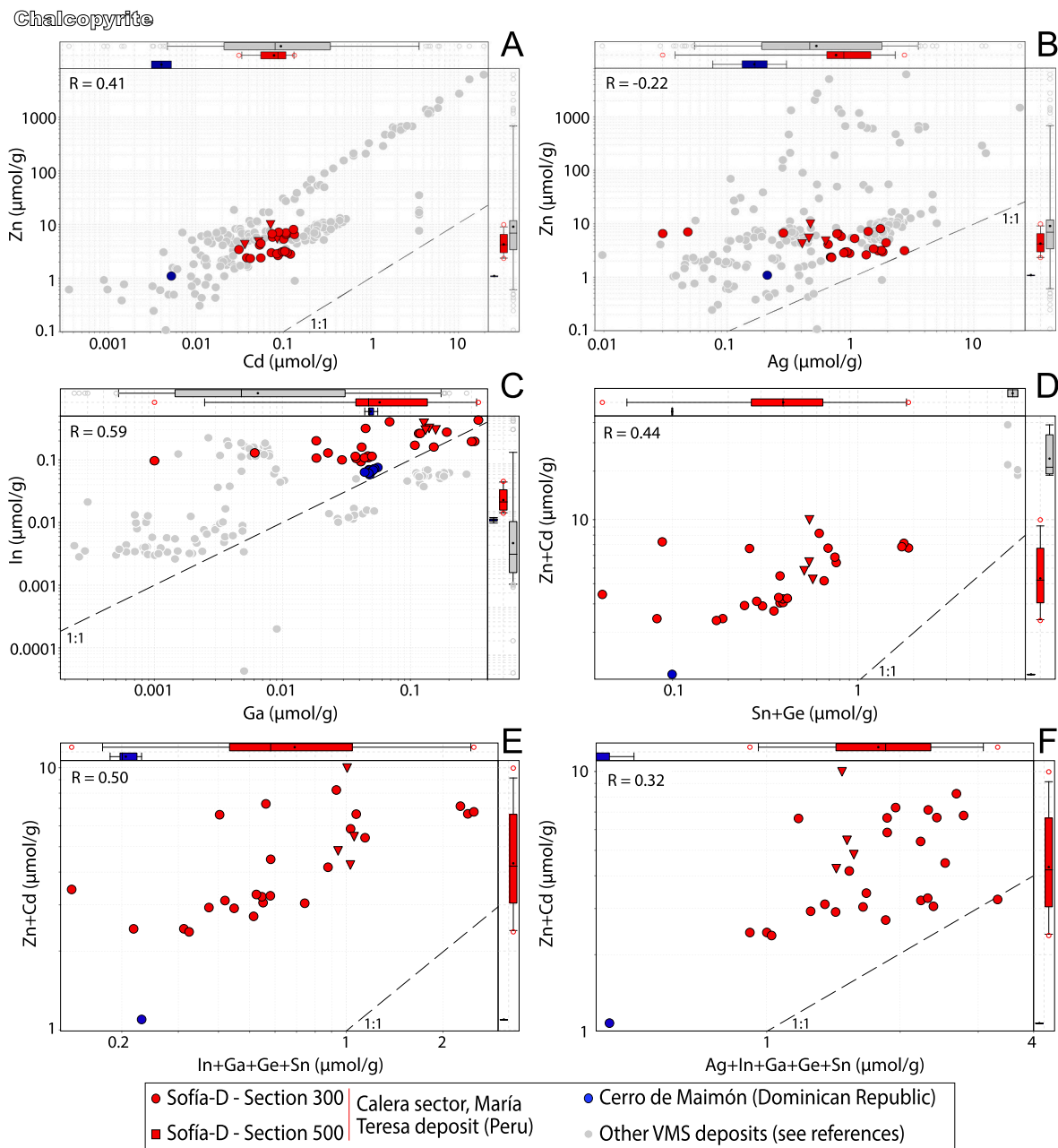


Fig. 11. Binary plots showing inter-element correlations in chalcopyrite: (A) Bi vs. Cu, (B) In vs. Cu, (C) Ga vs. Cu, and (D) Sn+In+Ge+Ga vs. Cu+Ag. Dashed lines represent 1:1 M correlation. Additionally, contents are presented in box-and-whisker plots, which provide IQR (box), 1.5 IQR (whiskers), median (line) and outliers (circles). The composition of sphalerite from other VMS deposits worldwide are shown for comparison (data from [Carvalho et al., 2018](#), [George et al., 2018](#), and [Revan et al., 2014](#)).

to 255–249 °C for sample 2019-MT-002 at 74.6 m; [Fig. 7J](#)). This decrease in the calculated crystallization temperature is actually a numerical translation of the depletion in In and Mn (directly proportional to the GGIMFIS temperature) and the enrichment in Ge (inversely proportional to the GGIMFIS temperature) bottom to top in sphalerite of the Sofia-D massive sulfide body ([Fig. 7](#)). Noteworthy, an increase in the availability of Cu in the mineralizing fluid towards the Cu zone probably also enhanced the incorporation of In to the sphalerite crystal lattice through the coupled substitution mentioned above ($\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$) during zone refining processes.

The Cd:Zn ratio in chalcopyrite is commonly used to trace variations in physicochemical conditions during its crystallization. Higher Cd:Zn ratios are typically ascribed to higher temperatures of crystallization ([George et al., 2018](#)). In the Sofia-D sulfide body, chalcopyrite shows

rather constant Cd:Zn ratios at different depths and sections ([Fig. 8K](#)) suggesting relatively homogeneous physicochemical conditions during the crystallization of this mineral throughout the Cu zone of the mineralized body ([Fig. 2](#)). In contrast, the increase of Se contents in chalcopyrite (and sphalerite) with depth would suggest an increase in the temperature and/or magmatic volatile influx ([Auclair et al., 1987](#); [Huston et al., 1995](#); [Layton-Matthews et al., 2008](#); [Martin et al., 2019](#)). The concentration of Se, and most trace elements indeed, in the structure of chalcopyrite is distinguishingly strongly influenced by the co-crystallizing phases so that the continuous crystallization of galena during zone refining probably partitioned Se and depleted chalcopyrite in this element ([Cook, 1996](#); [Carvalho et al., 2018](#); [George et al., 2018](#)). The progressively increasing abundance of galena towards the upper sections of the Sofia-D body may therefore explain the progressive

depletion of Se in chalcopyrite at shallower levels.

In short, our results suggest that in zone-refined VMS mineralization such as the Sofía-D massive sulfide body in the María Teresa deposit, chalcopyrite and particularly sphalerite register systematic changes in their trace element compositions across the mineralized body. Bottom to top of the sulfide body, observed trends include i) in sphalerite, depletion in In, Cu, Mn, Se, and Bi and enrichment in Ge; and ii) in chalcopyrite, depletion in Se and Bi (the latter probably in microinclusions). Furthermore, these patterns agree with expected physicochemical trends involving decreasing temperatures and influx of hydrothermal fluids with a direct magmatic contribution towards the margins of the sulfide body (Auclair et al., 1987; Huston et al., 1995; Layton-Matthews et al., 2008; Yeats et al., 2014; George et al., 2018; Schmidt et al., 2019; Martin et al., 2019).

Ore mineralization at the Sofía-D massive sulfide body in the María Teresa deposit underwent contact metamorphism connected to nearby granodiorite intrusions (Fig. 2). The identification of key trace-element distribution patterns in sphalerite and chalcopyrite across this body and their reconciliation with expected original trends suggest that contact metamorphism occurred in a closed system and more importantly, that body-scale trace element distribution was largely unaffected. Similarly, Polliand et al. (1999) determined that in the Perubar VMS deposit, sulfur isotope re-equilibration between sulfate-sulfide pairs during contact metamorphism occurred in a closed system and only between adjacent mineral grains. Lockington et al. (2014) found out that the content of lattice-bound trace elements in sphalerite (e.g., Mn, Fe, Cd, Cu, In, Ge, Hg, and Se) from metamorphosed VMS deposits worldwide showed no correlation with the metamorphic grade but were defined primarily by physicochemical conditions during initial deposition; in contrast, the contents of elements that typically occur as micro- and nano-inclusions in sphalerite (e.g., Pb and Bi) decreased with increasing metamorphic grade. Therefore, the use of trace element patterns defined by Mn, In, Ge, and Se in sphalerite, and of Se in chalcopyrite, are likely to reflect primary trends.

6. Conclusions and implications

In this study, systematic trace element determinations have been conducted in sphalerite and chalcopyrite from the Cretaceous arc-related Palma, Perubar, and María Teresa VMS deposits in Peru and the Cerro de Maimón VMS deposits in the Dominican Republic. Sphalerite from the four deposits incorporated Fe, Mn, Cd, Hg, Ag, Sb, Se, In, Ge, and Ga in solid solution. The incorporation of other elements such as Sn, Tl, Bi, and Pb possibly resulted in some cases from microinclusions as suggested by occasional ragged ablation spectra. The contents of minor and trace elements in analyzed sphalerite show significant variations even between samples from any individual deposit. Iron contents (44009–18,168 ppm) correlate negatively with Zn suggesting a dominant $\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$ simple isovalent substitution. Cadmium (7082–4972 ppm), Mn (4578–140 ppm), and Hg (453–107 ppm) are much concentrated than Pb (9.5–1.1 ppm), Sb (3.7–0.31 ppm), Tl (0.70–0.045 ppm), and Bi (0.60–0.0055 ppm).

Regarding potential *high-tech* by-products in the studied sphalerite grains, In (42–1.6 ppm, up to 415 ppm) and Ga (31–5.9 ppm, up to 96 ppm) contents are mostly higher than Ge contents (0.85–0.16 ppm, up to 9.3 ppm). In samples from the María Teresa deposit, the concentrations of In and Cu yield a positive correlation trend with a slope ~ 1 corroborating a $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ coupled substitution. The dataset also shows a positive correlation trend between Ga and Cu; analyses of sphalerite from the María Teresa, Perubar and Cerro de Maimón deposits lie close to the Ga:Cu 1:1 ratio line thus pointing to a $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ coupled substitution. Furthermore, the narrow arrangement of

most analyses along the 1:1 ratio line in a In + Ga + Ge + Sn vs. Cu + Ag plot suggests the occurrence of additional, more complex substitution mechanisms involving different combinations of monovalent (Cu^+ , Ag^+), divalent (Ge^{2+}), trivalent (In^{3+} , Ga^{3+} , $[\text{Sn}^{3+}]$), and quadrivalent (Ge^{4+} , Sn^{4+}) cations.

The trace element composition of chalcopyrite was investigated for the María Teresa and Cerro de Maimón deposits as chalcopyrite is a trace phase in the Palma and Perubar deposits. Trace elements such as Zn, Ag, Sn, Cd, Se, In, Ga, and Ge are present in solid solution as opposed to the likely occurrence of Mo, Au, Tl, Sb, Pb, and Bi in microinclusions. Deleterious elements show low contents (Zn = 426–190 ppm, Se = 64–22 ppm, Sn = 53–1.3 ppm, Cd = 12–6.0 ppm, Tl < 11 ppm, Bi < 10 ppm, Ni < 2.5 ppm, As < 2.0 ppm, Hg < 3.8 ppm) relative to reference maximum contents before charge is incurred *sensu* Fountain (2013). The contents of Ag (136–23 ppm, up to 296 ppm) and Au (<0.40 ppm) are also relatively low. In analyzed chalcopyrite grains, the contents of the potential high-tech by-products In (28–8.4 ppm, up to 49 ppm) and Ga (8.3–2.9 ppm, up to 24 ppm) are lower, and the contents of Ge (5.9–0.70 ppm, up to 80 ppm), higher, than in sphalerite from the respective deposits. General positive correlation trends between Zn, Cd, In, Ge, and Ga in chalcopyrite suggest varied coupled substitution mechanisms of Fe (+2 and +4) and Cu (+1 and +3) with fluctuating valences due to covalent bonding in this mineral.

A comprehensive sampling across the Sofía-D massive sulfide body in the María Teresa deposit revealed significant mineral and trace-element zonation. Mineral zones in Sofía-D include a barren lower part with dominant pyrite sheathed upwards by a Cu zone rich in chalcopyrite that in turn grades to sphalerite $\pm$ fahlore $\pm$ galena $\pm$ chalcopyrite (Zn zone) and galena + sphalerite $\pm$ fahlore $\pm$ barite (Pb – Zn – Ag zone) carapaces. Strong replacement of early sphalerite by chalcopyrite and the observed metal zonation strongly suggest zone refining. The contents of some trace elements in sphalerite and to a lesser extent in chalcopyrite display systematic vertical distribution patterns. In sphalerite, these include, bottom to top of the sulfide body, depletion in In, Cu, Mn, and Se and enrichment in Ge. This distribution agrees with the expected decrease of the crystallization temperature and/or volatile magmatic influx from the feeders of the massive mineralization. In chalcopyrite, the distributions of trace elements and Sc/Zn ratios are rather uneven except for constant enrichment in Bi (probably present as micro-inclusions) and Se towards the basal portion of the sulfide body. Unlike sphalerite, the trace element composition of chalcopyrite is strongly dependent on the co-crystallizing phases over physicochemical conditions. Co-crystallization and/or reequilibration of chalcopyrite with galena at shallower levels may therefore account for progressive depletion of Se in chalcopyrite.

The Sofía-D massive sulfide mineralization shows strong textural and mineralogical evidence of recrystallization during thermal metamorphism caused by nearby granodiorite intrusions. The impact of the thermal metamorphism in the absolute trace elements contents in sphalerite and chalcopyrite is unknown. However, the preservation of expected original trace element distribution patterns would suggest a closed metamorphic system and that element interdiffusion was local, probably attaining only contiguous sulfide grains. Accordingly, we propose that the distribution patterns described in this article for lattice-bound trace elements in sphalerite and chalcopyrite may be useful during exploration in order to trace the polarity of massive sulfide bodies in metamorphosed and tectonized terranes.

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Lisard Torró: Conceptualization, Investigation, Writing – Original Draft, Project Administration, Funding Acquisition

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Jean Vallance: Methodology, Investigation

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Brayam A. Ortiz-Benavente: Methodology, Investigation

Cyril Chelle-Michou: Validation, Writing – Review and Editing

Joaquín A. Proenza: Resources, Writing – Review and Editing, Supervision

Lluís Fontboté: Resources, Writing – Review and Editing, Supervision

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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